

Review

Enhancing tribological properties of MoS₂ coatings through doping: a review of mechanisms and applications

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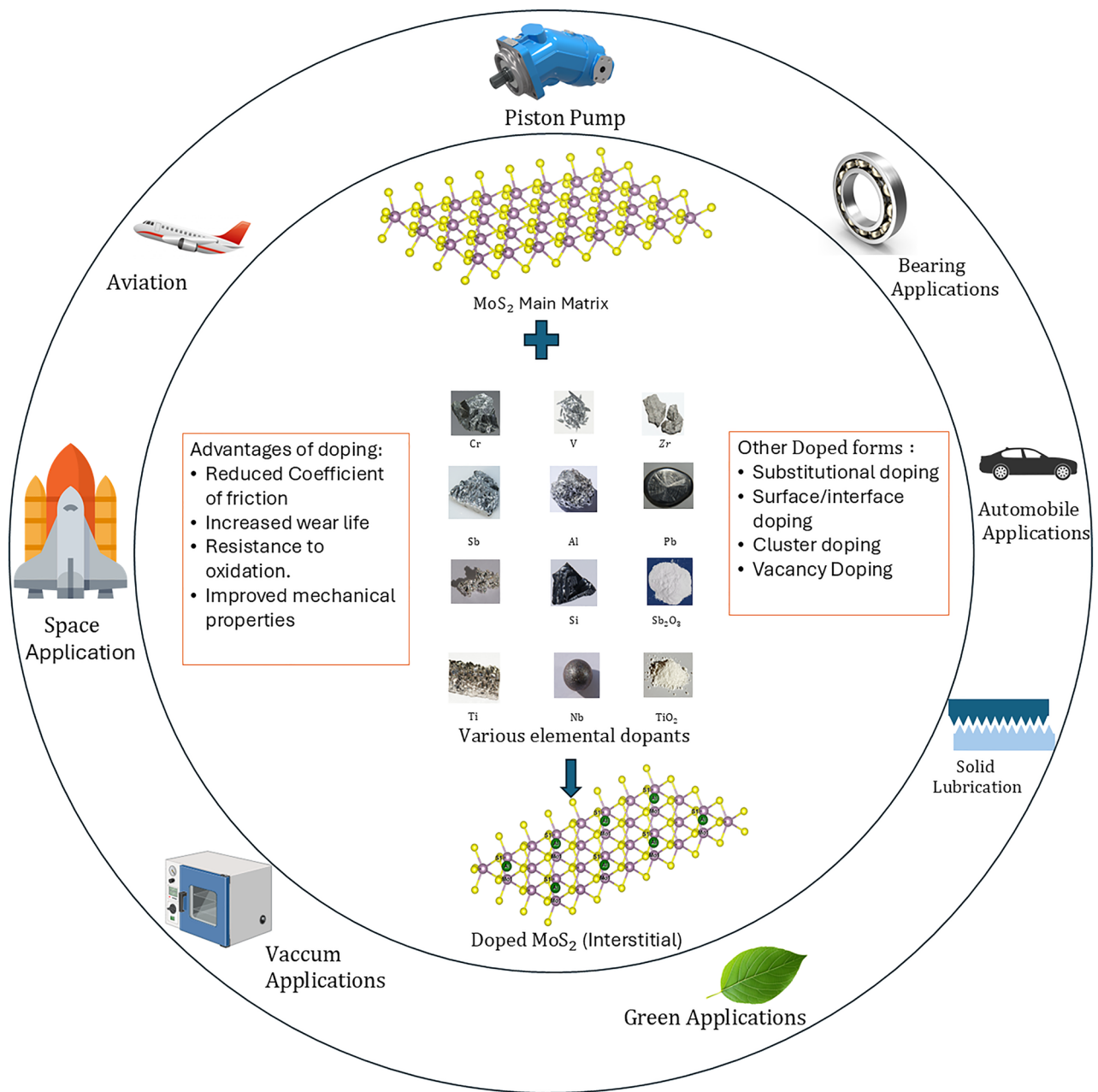
Abstract

MoS₂ is a very popular solid lubricant that has significant applications in the field of aerospace and automobiles. Due to its affinity towards oxygen and moisture, it has been modified through different mechanisms, such as doping and co-doping. This review comprehensively analyzes advanced materials used to alter MoS₂, highlighting their impact on wear resistance, mechanical durability, and environmental stability. Additionally, we discuss the latest fabrication and synthesis techniques tailored for tribological applications, emphasizing their role in improving coating performance. By elucidating the mechanisms through which dopants contribute to multifunctional MoS₂ coatings, this review offers valuable insights into their potential for next-generation lubrication technologies. The findings presented here serve as a foundation for future research to optimize MoS₂-based coatings for extreme environments and multifunctional applications.

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Graphical Abstract



Keywords Solid lubricant · Dopant · Tribology · Molybdenum disulfide · Transfer film

1 Introduction

Tribology, the study of friction, wear, and lubrication, is a field of immense scientific interest due to its wide range of applications in the field of engineering. It involves understanding the properties of existing lubricants, enhancing their effectiveness, and developing new lubrication methods to decrease friction coefficients, reduce wear rates, and improve component lifetimes. Due to the multifaceted interaction between moving surfaces, tribology must deal with various complex analyses. Despite their complexity, researchers have been challenged by studying these phenomena in isolation as many factors influence them, such as the structural, mechanical, and chemical properties of the surfaces involved. However, advancements in computational and experimental techniques have made tribology a vibrant and active field of inquiry, with research teams worldwide dedicated to understanding and leveraging its diverse aspects [1, 2]. A significant portion of a developed nation's gross domestic product is spent on mitigating the adverse effects of friction and wear, including the direct loss of Energy and the cost of replacing damaged machine components [3]. Therefore, tribology research focuses on optimizing losses due to high wear and friction.

Molybdenum disulfide (MoS_2) is a transition metal dichalcogenide (TMD) with an indirect bandgap of ~ 1.2 eV in its bulk form and a direct bandgap of ~ 1.8 eV when the sample is reduced to a single or few monolayers. The shift from an indirect to a direct bandgap when moving from the bulk to monolayer form, combined with additional characteristics such as spin coupling and significant exciton binding energy, makes MoS_2 a compelling topic for scientific investigation. Its layered structure provides low friction and wear owing to the easily sheared planes, and low-friction applications are highly lubricating under vacuum and low humidity conditions [4]. Owing to these properties, MoS_2 is employed as a solid lubricant and is very effective in space, automotive, and manufacturing applications, where the reduction of friction and wear is of primary importance for elongating the lifespan and enhancing efficiency. However, this excellent performance has several drawbacks, including the oxidation of MoS_2 when exposed to humid atmospheres and the loss of efficiency during high loads and extreme temperatures [5, 6].

Researchers have focused on modifying MoS_2 with various metals, semiconductors, ceramics, and alloys to overcome these drawbacks. Doping introduces atoms into the MoS_2 structure, changing its chemical and mechanical properties, such as hardness, wear resistance, and oxidation resistance. Dopants can improve MoS_2 's tribological performance by altering its crystalline structure or creating composite materials with enhanced durability. For example, the incorporation of elements like titanium (Ti) [7], chromium (Cr) [8], and lead (Pb) [9] has been shown to improve the hardness and wear resistance of MoS_2 coatings. Meanwhile, soft materials like gold (Au) [10] or antimony (Sb) [11] can reduce friction, enabling the lubricant to perform better in a wide range of operational conditions.

Over the past two decades, numerous studies have highlighted the advantages of MoS_2 -metal composite and multi-layered coatings deposited through co-sputtering methods. These coatings have shown enhanced hardness, adhesion strength, and tribological performance in ambient environments compared to pure MoS_2 coatings [12–23]. Among these, MoS_2 -Ti composite coatings, branded as MoST™ by Teer Coatings Ltd, have been particularly well-studied [12–18]. In various industrial applications, MoST™ coatings have significantly improved performance over traditional hard coatings, especially in cutting and forming tools under normal operating speeds. Similarly, MoS_2 -Cr composite coatings have also garnered considerable interest [19–23]. Chromium is anticipated to function comparably to titanium in MoS_2 -metal composites, and in specialized applications such as machining brass and stamping copper-based coinage, Cr-based MoS_2 coatings are expected to offer superior performance [13].

Given the extensive research on MoS_2 and its doped variants, this review aims to provide a comprehensive overview of the recent advancements in doping MoS_2 to enhance its tribological properties. While numerous studies have focused on specific dopants or applications, a critical review that consolidates these findings and discusses emerging trends is necessary. This review will cover the mechanisms through which doping improves tribological performance and highlight key dopants and their impact on MoS_2 's performance under various operational conditions. At the end of the article, this review gives an insight into the different research domains regarding modifying MoS_2 that have not been explored and suggests materials with huge potential as dopants. Ultimately, this review is crucial for guiding future research and industrial applications in developing more efficient and durable solid lubricants.

Polytypes: 1T, 2H and 3R

Space group: $P\bar{3}m1$, $P6_3/mmc$ and $R3m$

Mo-S coordination: Octahedral, Trigonal prismatic and Trigonal prismatic

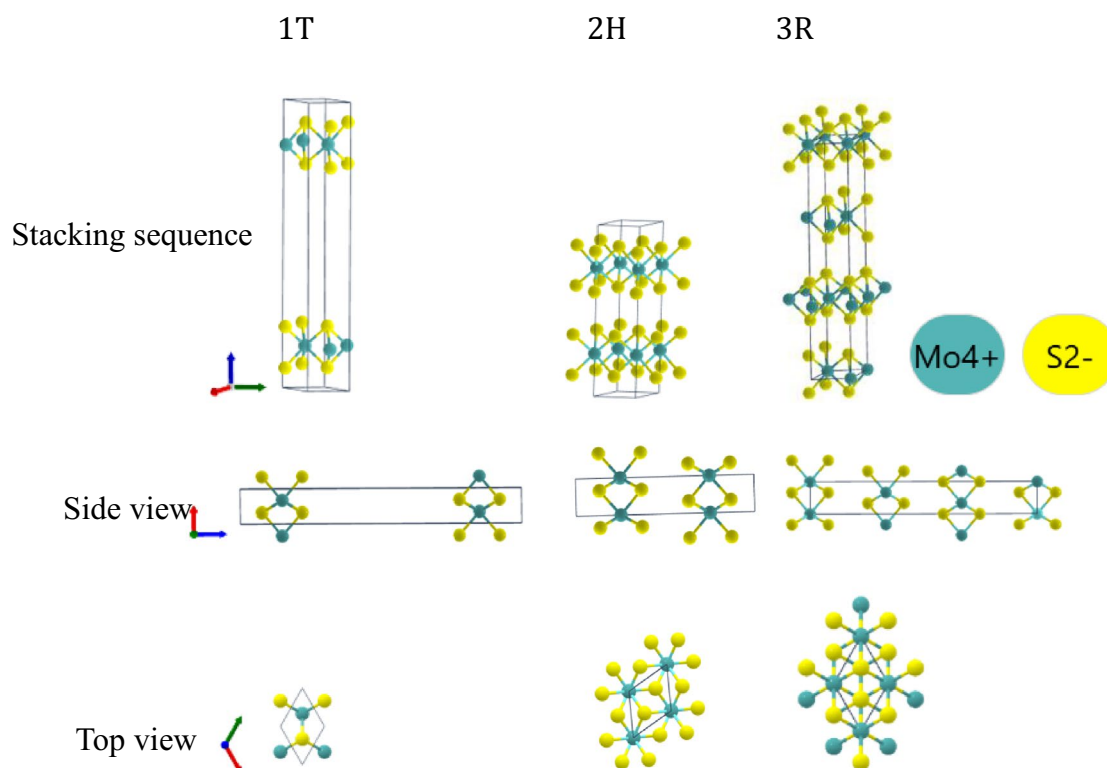


Fig. 1 Three typical phases of bulk MoS_2 crystals based on known polytypes

2 Structure and synthesis of MoS_2

2.1 Structure

MoS_2 , a 2D transition metal dichalcogenide (TMD), features a layered structure with Mo atoms sandwiched between S atoms. Weak van der Waals forces between layers enable easy exfoliation and contribute to its solid lubrication properties by resisting shear forces. Its structure is classified into three crystalline phases: 2H, 1T, and 3R (Fig. 1) [24, 25].

The 2H phase is the most stable and naturally occurring structure of MoS_2 , featuring a hexagonal lattice with a trigonal prismatic arrangement, where Mo atoms coordinate with six S atoms in an ABA stacking sequence. Its point group is $D6h$ [24, 26, 27]. Bulk 2H- MoS_2 has an indirect bandgap of ~ 1.2 eV, while the monolayer form, lacking interlayer coupling, exhibits a direct bandgap of ~ 1.9 eV [28, 29]. This transition enhances photoluminescence, making monolayer MoS_2 ideal for optoelectronic applications like photodetectors and LEDs [25].

The metastable 1T phase of MoS_2 exhibits metallic properties due to its octahedral coordination and ABC stacking, distinguishing it from the hexagonal 2H phase. It features stronger in-plane bonds, weaker interlayer interactions, and enhanced charge carrier mobility [28]. Partially filled Mo d-orbitals contribute to its conductivity [30]. The 1T phase can be induced from 2H via chemical treatments like Li intercalation, which restructures the layers. Its superior charge transfer capabilities make it promising for energy storage applications, including supercapacitors and battery electrodes.

The 3R phase of MoS_2 is a rare, metastable form with a rhombohedral structure and three-layer periodicity, typically synthesized under specific conditions like high pressure or temperature. It differs from the 2H phase by a shifted stacking order, breaking inversion symmetry and enhancing nonlinear optical properties. However, it reverts to the 2H phase

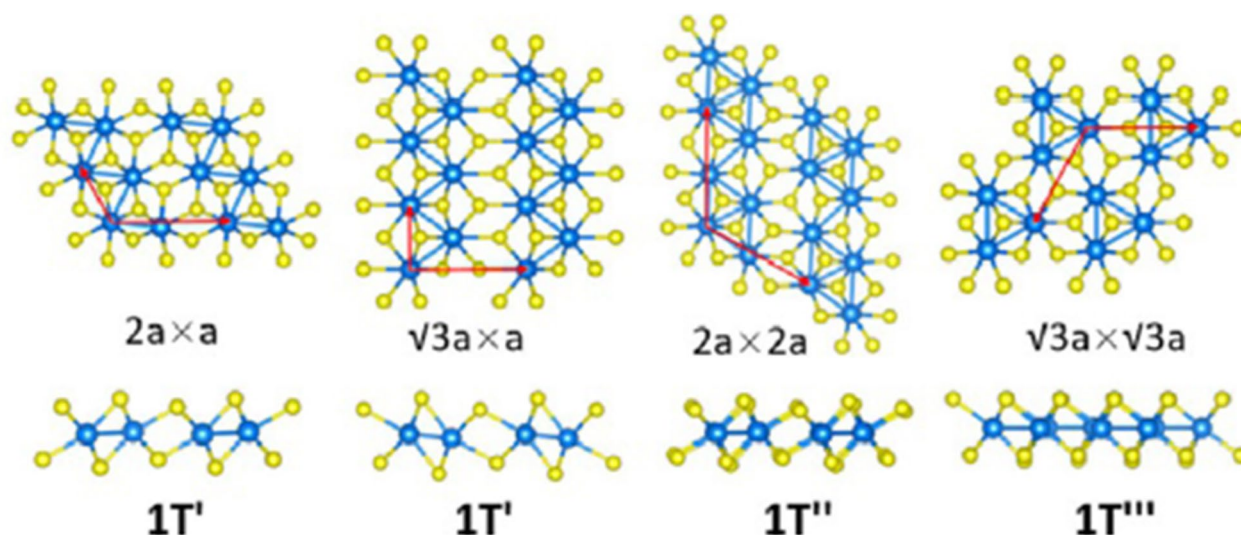


Fig. 2 $1T'$, $1T''$ and $1T'''$ phases of MoS₂. This was reproduced with permission from Ref. [25], Copyright (2018), Chem. Eur. J

upon thermal treatment. The coexistence of 2H, 1T, $1T'$, and 3R phases highlights MoS₂'s structural diversity, offering versatility for research and applications [26, 29].

The $1T'$ and $1T''$ phases are distorted derivatives of the 1T structure, arising from slight atomic displacements in the octahedral arrangement (Fig. 2). These polymorphs exhibit unique electronic properties, ranging from semiconducting to semi-metallic behavior. The $1T'$ phase, with zigzag Mo chains, is mainly explored for quantum spin Hall effects and topological insulator studies. Though less stable than the 2H phase, they often appear in chemically exfoliated or doped MoS₂. Advanced techniques like Raman spectroscopy and XRD are crucial for identifying these phases through vibrational shifts and symmetry changes [24–26].

Heteroatom doping influences MoS₂ phase transitions and properties [31]. Elements like Co, Mg, Ni, and Sb modify their electronic and structural characteristics, enhancing catalytic activity. Co and Ni doping, for example, induce the 2H-to-1T transition by electron injection, stabilizing the metallic phase and increasing active sites for reactions like hydrogen evolution reaction (HER) [32]. Such modifications optimize MoS₂ for energy conversion, improving charge transfer and catalytic efficiency.

In summary, the structural versatility of MoS₂, from its stable 2H phase to the metastable 1T and 3R phases, coupled with the tunability through doping and phase transitions, underscores its potential in a wide array of applications. The ability to manipulate its phase composition and structure allows researchers to tailor MoS₂ for specific needs, ranging from electronic and optoelectronic devices to energy storage and conversion technologies. Understanding the relationship between the structural properties of MoS₂ and its resulting electronic behaviors remains a critical area of research, offering insights into the design of next-generation materials based on this promising 2D material.

2.2 Synthesis

MoS₂ is a very popular material with diverse applications in tribology, electronics, biosensors, etc. Different applications require different synthesis techniques. As our review focuses on MoS₂'s tribological application, we focus on the methods used to synthesize MoS₂ for tribological applications. Table 1 gives an overview of different techniques used by tribologists.

Recent advancements in MoS₂-based coatings have improved tribological properties for high-performance applications. This section focuses on fabrication techniques, particularly PVD, which deposits MoS₂ and dopants in a vacuum chamber to form uniform films. Optimizing key parameters like temperature and deposition rate enhances coating performance, with metal dopants reducing friction and improving wear and oxidation resistance.

Table 1 Various synthesis and fabrication techniques used for MoS₂

Synthesis method/sub type	Literature	Precursor	Synthesis condition	Substrate	Surface morphology
CVD					
	[33]	S, YCl ₃ ·6H ₂ O, MoO ₃	Furnace pressure, 1.6 Pa 20 °C/min until 700 °C and then 5 °C/min till 820 °C for 2 h in Ar:H ₂ = (100:10, sccm) environment	TiN coated cemented carbide blades	Disordered MoS ₂ flakes and S particles wrapped in little MoS ₂ flakes
	[34]	Sb, S, MoO ₃ , Ag	Temperature: 820 °C, furnace pressure: 6 kPa, time: 4 h	TiN-coated cutters	Flower-like microstructure
	[35]	ReO ₃ ; MoO ₃ ; H ₂ S; H ₂	Temperature: Varies upto 800 °C	Porous TiO ₂	Smooth surface with cracks
PVD					
Unbalanced magnetron sputtering	[36]	MoS ₂ ; Ce-Ti alloy (1:1 ratio) (10 min Ce, 0.6 Pa, 100 W)	Pressure: 0.6–1.8 Pa; temperature: 200 °C; Ar flow (50 sccm); time: 60 min	9Cr18 steel	Porous structure
Hybrid: HIPIMS-DCMS, MFMS, linear anode layer ion	[37]	C-Si, MoS ₂ , Cr	Pressure: 3 × 10 ⁻³ Pa, time: 450 min	Si (100), AISI 304 stainless steel	Columnar
High target utilisation sputtering	[38]	MoS ₂ , Cr	Ar (100 sccm), N ₂ (5–25 sccm), pressure: 0.73–0.95 Pa, time: 35–100 min	Si (100), AISI 316 steel	Featureless
Magnetron sputtering	[39]	MoS ₂ , B	Pressure: 5 × 10 ⁻⁴ Pa, Ar = 80 sccm at 200 °C, time: 2 h	Si (100)	Dense granular microstructure
DC sputter deposition with electron beam assist	[40]	No description	No description	Stainless steel (440C)	Columnar microstructure
Unbalanced magnetron sputtering	[9]	MoS ₂ , Pb, Ti, and Ta	Ar (16 sccm), time: 30–200 min	Si (100)	Dense and smooth
Unbalanced magnetron sputter ion plating system	[41]	MoS ₂ , Ti, LaF ₃	Pressure: 5 × 10 ⁻⁵ Torr, 20 min; Ar (30 sccm); substrate rotation: 4 rpm	GCr15 bearing steel and Si	Columnar crystal microstructure
Magnetron sputtering	[42]	MoS ₂ , Cu	Pressure: 0.8 Pa; Ar (80 sccm); substrate rotation and temperature: 10 rpm, 200 °C, time: 70 min	Si (100)	Turbulence-like MoS ₂
Unbalanced magnetron sputtering	[43]	MoS ₂ , Ti, Si	Base pressure: < 1.0 × 10 ⁻³ Pa; time: 2 h, environment: Ar	Stainless steel (304), P-type Si (100)	Multilayer columnar nanostructure
Closed field unbalanced magnetron sputtering	[44]	MoS ₂ , Zr, Ti	Base pressure: < 1.33 × 10 ⁻³ Pa, 30; current: 0.5–1.5 A	Si, 304 stainless steel	Cauliflower-like
Closed field magnetron sputtering	[45]	MoS ₂ , Si, Ti	Si (0–0.6 A)	Si (100), 304 stainless steel	Multilayer nanostructure
Closed-field unbalanced magnetron sputtering	[46]	MoS ₂ , Cr, Nb, Ti, Al, V	Base pressure: 3.0 × 10 ⁻³ Pa; sputtering pressure: 0.3 Pa; Ar (70 sccm)	Si (100)	Columnar crystal structure

Table 1 (continued)

Synthesis method/sub type	Literature	Precursor	Synthesis condition	Substrate	Surface morphology
Magnetron sputtering	[47]	MoS ₂ , Ti, Cr	Base pressure: 1.5×10^{-3} Pa, Sputtering pressure: 0.5 Pa; Ar (35 sccm)	Ti-6Al-4 V alloy	Broccoli and hemisphere-like
DC magnetron sputtering	[48]	MoS ₂ , ZnO, Ti, Al	Base pressure: 5×10^{-4} Pa; sputtering pressure: 1 Pa; time: 3 h	SUS 201 steel, Si	Columnar structure
HiTUS	[49]	MoS ₂ , N ₂ , Ar	0–40 at.% [7]	M2 steel discs	Dense microstructure
Magnetron sputtering and high-energy ion bombardment plasma beam assisted deposition	[50]	MoS ₂ , Ag	Pressure: 9.5×10^{-2} Pa, 120 min, Ar, f Kauffmann ion sources: 75 W	Si	Dense tribofilm with embedded Ag nanoparticles
Magnetron sputtering	[51]	MoS ₂ , Pb, Ti	Not given	304 Stainless Steel	Cauliflower, smooth and dense
Magnetron sputtering	[52]	MoS ₂ , Ti, graphite	Base pressure: 1.5×10^{-3} ; sputtering pressure: 0.65 Pa; Pa, temperature: 150 °C	Si (100)	Compacted granular structure
Hydro-thermal	[53]	Na ₂ MoO ₄ ·2H ₂ O, CH ₃ CSNH ₂ , CO(NH ₂) ₂	200 °C, 24 h	Willow catkins (template)	Loosely packed MoS ₂ nano-sheets
	[54]	Na ₂ MoO ₄ ·2H ₂ O, TAA, GO, N ₂ H ₄ ·H ₂ O	100 mL at 200 °C and 230 °C for 24 h	N/A	Nanosheets
	[55]	h-BN (h-BN-OH), (NH ₄) ₂ MoO ₄ , CH ₄ N ₂ S	210 °C for 24 h	Epoxy (EP) composite coating	Nanosheets
Dispersion > dip-spin coating	[56]	g-C ₃ N ₄ , Na ₂ MoO ₄ , CH ₄ N ₂ S, C ₆ H ₈ O ₇ , DI water	190 °C for 24 h	AISI 52100 steel	MoS ₂ nanosheets intertwined with CN sheets
Reflux	[57]	Everlube 620C, Lube-Lok 5306 MoS ₂ , Sb ₂ O ₃ , PbHPO ₃	As purchased	304L stainless steel	Smooth tribofilms
	[58]	MoS ₂ , NiCl ₂ ·6H ₂ O, DI, H ₂ SO ₄	Ultrasonication (90 min), reflux temperature: 80 °C, time: 12 h	N/A	Nanosheets
Spark plasma sintering (SPS)	[59]	MoS ₂ , BaSO ₄ , Al ₂ O ₃ , BaMoO ₄	Temperature: 1150 °C and pressure: 25 MPa; heating rate is 250 °C/min with holding time 4 min	N/A	Multilayer structure
In situ chemical polymerization	[60]	Polyaniline, MoS ₂ , HCl	Room temperature; several hours	Steel discs (AISI 52100), Cu disc	Layered structure
Solution blending/spray coating	[61]	Ti ₃ AlC ₂ , MoS ₂ , ethanol	Sonication: 5 min; environment: dry N ₂	AISI 52100 steel	Multilayered structure with overlapping nanosheets
Laser cladding	[62]	Cu-10Al powder, Ni packed MoS ₂	1200 W, 2.5 mm/s, 4 mm, 50%, 11.5 g/min, Ar (10 L/min)	Q235 steel	Pores or cracks
	[63]	NiCrBSi, Ag, MoS ₂	Laser power density: 65–80 W/mm ² , Ar atmosphere	1.4301 grade stainless steel	Ni dendrites surrounded by Cr ₂ S ₃ and scattered S pockets

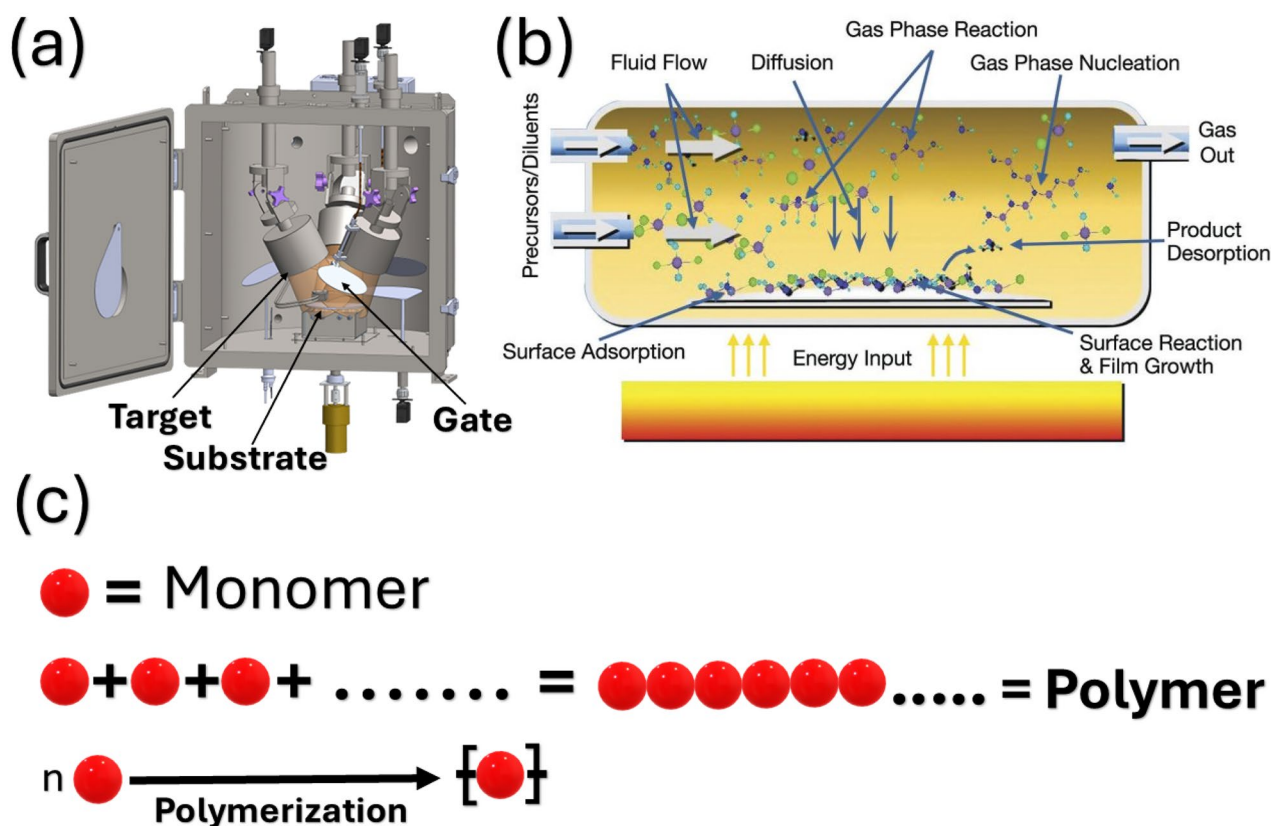


Fig. 3 Various synthesis methods for MoS₂ thin films. **a** Schematic representation of the physical vapor deposition (PVD) machine using DC magnetron sputtering. **b** Overview of chemical vapor deposition (CVD), illustrating key processes such as precursor flow, gas-phase reactions, surface adsorption, film growth, and product desorption. **c** The polymerization process depicts the conversion of monomers into polymer chains through a stepwise reaction mechanism

2.2.1 Sputtering

Sputtering remains the preferred method for doping MoS₂ in tribological applications due to its ability to produce dense, wear-resistant, low-friction coatings. It enables fine microstructure control, supports multilayer deposition, and enhances oxidation resistance, which is crucial for high-humidity environments. Cr-, Ti-, and Zr-doped MoS₂ coatings via sputtering exhibit stable friction and durability. HIPIMS further improves wear resistance in amorphous MoS₂-Ti films. Unlike alternative methods like CVD and laser doping, which face scalability and temperature challenges, sputtering offers a cost-effective, scalable solution for optimizing MoS₂-based coatings in tribology.

Sputtering encompasses diverse techniques, including planar diode gate discharge, triode, magnetron, and ion beam sputtering, along with advanced variations, such as biased target ion beam deposition and specialized magnetron methods (balanced, unbalanced, closed-field, reactive, pulsed, and high-power impulses), which are designed to optimize film characteristics and meet specific deposition needs [64]. Figure 3 shows the process of physical vapor deposition, chemical vapor deposition, and polymerization.

2.2.1.1 Balanced magnetron sputtering Balanced magnetron sputtering modifies MoS₂ films by adjusting target power and current, influencing structural and tribological properties. For example, Sun et al. [42] doped MoS₂ with B and Cu, improving hardness, wear resistance, and surface compactness. Incorporating B content causes the wear rate to decrease from $5.66 \times 10^{-5} \text{ mm}^3/\text{Nm}$ for the pure MoS₂ to $5.5 \times 10^{-7} \text{ mm}^3/\text{N m}$ for the Mo-S-B film with 23.4 at.% B [39]. Similarly, Cao et al. co-sputtered MoS₂ with ZnO, Ti, and Al, forming layered structures with enhanced tribological performance [48]. The MoS₂ films were modified using a three-target magnetron sputtering system, where MoS₂ was co-sputtered with Ti for Ti-doping (MoS-T), while a graphite target was used for C/Ti co-doped (MoS-C-T) MoS₂ films. During the process, a controlled RF power (250 W) was applied to both the MoS₂ and graphite targets, while Ti doping occurred via

direct current (0.11 A) application. This setup created films with a dense, layered structure, enhancing their tribological and irradiation-resistant properties, as evidenced by the stable performance of MoS₂ crystallites under ion irradiation [52]. Yang et al., modified MoS₂ to deposit TiCrN/MoS₂-TiCr coatings using magnetron sputtering on a Ti-6Al-4 V substrate. A layer of MoS₂ doped with varying amounts of Ti and Cr was applied to the TiCrN coating to enhance its tribological properties [47].

2.2.1.2 Unbalanced magnetron sputtering (UBMS) UBMS enables MoS₂ doping with soft and hard phases like Pb and Ti for balanced lubricity and strength [9]. Pb doping reduces friction via PbS formation, while Ti improves hardness. LaF₃-doped Ti-MoS₂ films showed enhanced wear resistance at optimal concentrations [41]. Ce-Ti/MoS₂ coatings optimized at low pressures exhibited improved friction and wear properties [36]. MoS₂/Ti-MoS₂/Si multilayers enhanced durability under high loads [43].

2.2.1.3 Closed-field unbalanced magnetron sputtering (CFUBMS) CFUBMS allows precise MoS₂ doping, improving tribological properties in humid environments. Si doping densifies coatings, reducing oxidation [45]. Multi-target setups enable controlled doping gradients, influencing hardness and wear resistance [46]. Zr-doped MoS₂ transitioned from crystalline to amorphous, enhancing oxidation resistance [44].

2.2.1.4 Other sputtering methods HiPIMS and HiTUS create dense MoS₂ coatings with high adhesion. HiTUS-based N-doped MoS₂ showed enhanced hardness and wear resistance [38, 49]. Hybrid systems, such as HiPIMS-DCMS, improved film stability [37]. Ag-doped MoS₂ with ion bombardment enhanced friction reduction and wear resistance [50].

2.2.2 Chemical vapor deposition

CVD enables uniform dopant integration into MoS₂ coatings, enhancing tribological properties. Yi et al. [33] synthesized Y-doped MoS₂ using YCl₃ and MoO₃ in a tube furnace at 820 °C, achieving improved hardness and adhesion. Similarly, Sb/Ag-doped MoS₂, synthesized under similar conditions, exhibited a flower-like structure and enhanced lubrication [34]. Shalom et al. [35, 64] controlled Re doping in MoS₂, preventing segregation and improving wear resistance.

2.2.3 Sintering-based processes

Laser cladding enhances MoS₂ coatings by melting MoS₂-containing materials onto substrates. Wang et al. [62] encapsulated MoS₂ in Ni for Cu-10Al coatings, improving wear resistance. Torres et al. [63] used NiCrBSi-Ag/MoS₂ coatings, enhancing load-bearing and lubrication at high temperatures. Macknoja et al. [61] spray-deposited Ti₃C₂T_x/MoS₂ onto steel, achieving superior tribological performance. Rajendhran et al. [58] synthesized Ni-MoS₂ nanosheets via reflux, enhancing dispersion and wear resistance. Song et al. [59] developed Al₂O₃/MoS₂-BaSO₄ composites using spark plasma sintering, forming self-lubricating BaMoO₄ phases.

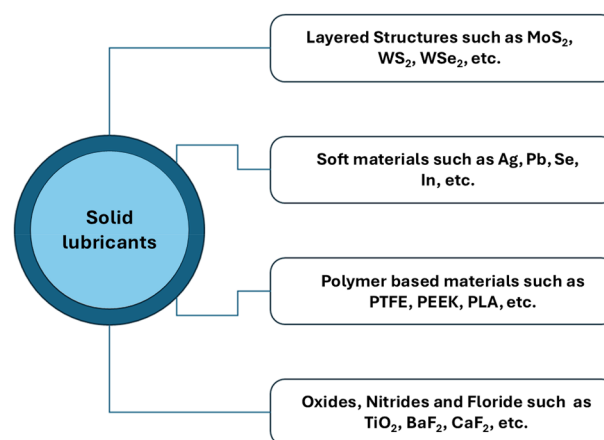
2.2.4 Solution based processes

Solution-based methods offer versatile and scalable approaches for doping molybdenum disulfide (MoS₂) to enhance its tribological properties. These techniques enable precise control over dopant distribution and concentration, which is crucial for tailoring the material's performance in various applications [65].

2.2.4.1 Hydrothermal process Hydrothermal synthesis produces MoS₂ with diverse morphologies, optimizing tribological properties. Parameters like temperature (180–230 °C) and surfactants (e.g., CTAB) control crystal growth [66, 67]. Feng et al. [60] synthesized MoS₂/PANI composites, improving conductivity for supercapacitors. Hydrothermally grown MoS₂ nanosheets enhance lubrication and wear resistance, forming tribological films under heavy loads [54, 67]. Doping MoS₂ with elements like nitrogen or graphene oxide further improves hardness, stability, and adhesion, making it suitable for demanding tribological applications [55, 68].

2.2.4.2 Wet chemical doping This method involves dissolving dopant precursors, such as transition metal salts, into a solvent to create a homogeneous solution. MoS₂ is introduced into this solution, allowing the dopant ions to intercalate

Fig. 4 Different types of solid lubricants (Information collected from [74])



or substitute into the MoS_2 lattice. For instance, incorporating oxygen into MoS_2 through controlled exposure has been shown to reduce friction and wear, as oxygen can cause amorphization of the as-deposited coatings without preventing the formation of a lubricious tribolayer [69].

3 Solid lubricant and its classification

When considering lubrication in industrial settings, liquid lubricants like oils and greases are often the first to come to mind [70]. These substances are commonly used in mechanical systems because of their better tribological properties like low wear, ease of replenishment, and simple application to various surfaces. However, certain operating conditions necessitate using solid lubricants instead [71], particularly in aerospace and space applications, where low temperatures make liquid lubricants too viscous or solid, rendering them ineffective [72, 73]. The development of solid lubricants saw significant Progress in the latter half of the twentieth century, driven by the demands of the space age.

Solid lubricants are also essential in dry machining operations, as they reduce wear on machine tools and lower the costs associated with using large quantities of liquid lubricants. Industries like food processing and textiles rely on solid lubricants to avoid contamination. Liquid lubricants could pose health hazards [4]. Overall, solid lubricants are suitable for extreme conditions. Figure 4 shows the different types of popular solid lubricants.

3.1 MoS_2 as a solid lubricant

Molybdenum disulfide (MoS_2) is a significant solid lubricant currently used, dating back to the previous century [75, 76]. It is a lamellar solid material composed of atomically thin planes that can easily slide against each other. Like graphite, another lamellar material, MoS_2 can function as a dry lubricant on its own, as an additive in oils or greases, or as part of a composite coating [1]. Unlike graphite, MoS_2 does not need assistance from humidity to perform well; its lubricative properties improve significantly in oxygen-deficient environments [77].

MoS_2 's reliability across extreme temperatures and vacuum conditions makes it ideal for aerospace and space applications. It serves as a solid lubricant in satellites and space probes, where liquid lubricants would freeze or evaporate. Since the 1960s, extensive research has focused on its synthesis, properties, and applications as a solid lubricant [78, 79].

Despite its long research history, efforts continue to enhance MoS_2 's lubricative properties. Recent advancements focus on incorporating MoS_2 into (nano-)composite coatings and improving its performance through controlled doping, where specific elements are introduced to modify its properties systematically [80, 81]. Developing multifunctional MoS_2 coatings in combination with different dopants and materials is essential for the more diverse application of MoS_2 as a solid lubricant. Figure 5 shows the significant progress in MoS_2 doping in the recent years.

The following section focuses on developing composite MoS_2 coatings using various dopants. It will discuss the development of doped MoS_2 over the last few decades and how the addition of dopants improved its tribological properties.

For several decades, dopants have been utilized to enhance the tribological properties of MoS_2 coatings [83]. These dopant atoms can be incorporated into the layered structure through various mechanisms, including substituting atoms

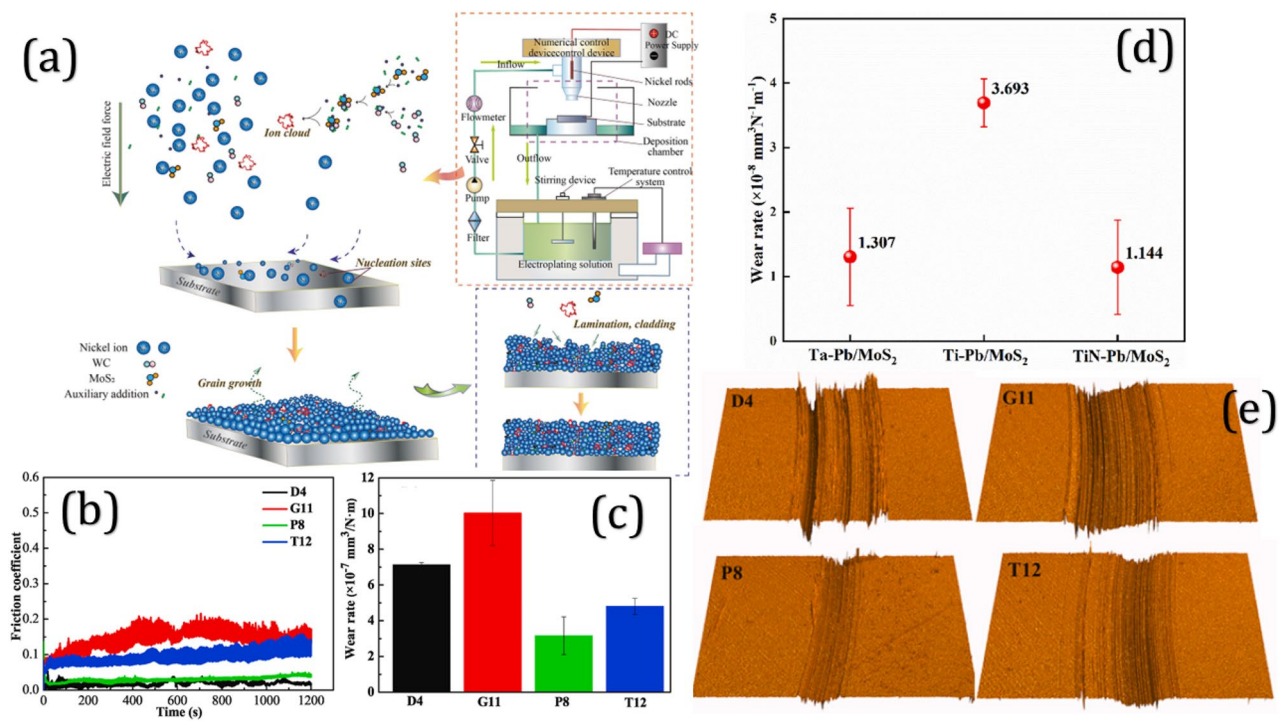


Fig. 5 A summary of different studies using various dopants to enhance the tribological properties of MoS₂ coatings. **a** Deposition of Ni-WC on MoS₂ coatings using physical vapor deposition to achieve improved tribological properties of MoS₂. **b** Tribological test analysis of different samples. samples D4, G11, P8, and T12 refer to specific MoS₂-(Cr, Nb, Ti, Al, V) composite coatings selected based on their friction coefficients. These samples represented the largest differences in friction performance among the 236 prepared samples. **c** Comparison of wear rate of four different samples. **d** Comparison of wear rate for Tantalum-Lead, Titanium-Lead, and Titanium Nitrate-Lead doped MoS₂ coatings. **e** Wear track of four different samples. (Reused with permission from Elsevier [9, 46, 82])

in the lattice, occupying interstitial sites, intercalating between layers, or existing as a separate phase dispersed in the material. The specific mechanism depends on the dopant type, concentration, and deposition process used to create the coating. Various deposition methods are employed to produce doped coatings. Sputter deposition, also known as Sputtering, was one of the earliest techniques and remains widely used. Other methods used for doped MoS₂ coatings include unbalanced magnetron sputtering, ion beam deposition, and pulsed laser deposition. Several dopants, including Ni [40], Cr [46, 84, 85], Ti [86], Au [87], Zr [44], and Sb₂O₃ [87], have been studied and demonstrated to enhance the tribological properties of MoS₂. Doped MoS₂ coatings have shown improved tribological properties, such as reduced friction, wear, and increased lifespan, compared to undoped MoS₂, particularly in Vacuum. Additionally, doped coatings exhibit superior oxidation resistance, enhancing their performance and storage capabilities in air [81].

4 Doping and Co-doping of MoS₂

Doping and Co-doping MoS₂ with different dopants can improve tribological performance, mainly wear resistance [87–90]. One frequently studied co-doping combination is Sb₂O₃/Au-doped MoS₂. Coatings with an optimal composition of 11% Sb₂O₃ and 7% Au have shown excellent tribological performance, with an order of magnitude lower wear rate than Sb₂O₃-doped MoS₂ at room temperature. This feature is crucial for space applications where endurance is paramount, leading to using Sb₂O₃/Au coatings in some space components [87].

Several studies have compared multiple dopants. In one study, measurements of friction and life of Ni, Fe, Au, and Sb₂O₃-doped coatings showed all to be better than undoped MoS₂, with Sb₂O₃/Au being the most effective [89]. Another comparative study of friction and endurance involving Al, Pt, Ag, W, Cr, Co, Ni, and Ti showed that some dopants did not improve performance while others had a positive effect. Optimal metal content was 5–8% for those that improved performance [20]. In comparing Au, Ti, Cr, and WSe₂, WSe₂ exhibited the lowest friction and wear, followed by Au, Ti, and then Cr [19].

Temperature dependence studies have shown exciting results as well. Friction and wear measurements from -80 to 180 °C for $\text{Sb}_2\text{O}_3/\text{Au}$, $\text{Sb}_2\text{O}_3/\text{C}$, Sb_2O_3 , Ti, and Ni coatings revealed that $\text{Sb}_2\text{O}_3/\text{Au}$ had the lowest friction above -25 °C, while Ni had the lowest friction at -80 °C [88]. $\text{Sb}_2\text{O}_3/\text{Au}$ also exhibited the lowest wear rate at room temperature. As the temperature increased, the Coefficient of friction reduced and vice versa, indicating that high room-temperature wear corresponded to a minor increase in friction with decreasing temperature [88].

Comparing different deposition methods for dopants can be challenging, as they can affect the properties of the resulting coating. However, one study compared undoped MoS_2 , Ti-doped, and $\text{Sb}_2\text{O}_3/\text{Au}$ -doped coatings deposited similarly. The study found that MoS_2 performed best at 30 °C and worst at 100 °C, $\text{Sb}_2\text{O}_3/\text{Au}$ was highly abrasive at 30 °C but performed well at 100 °C, and Ti exhibited good tribological behavior at both temperatures [90].

In summary, dopants consistently improve the tribological performance of MoS_2 coatings. These dopants can be broadly categorized into hard materials (such as Ti, Ni, and Sb_2O_3) and soft materials (like Au). Regarding oxidation resistance, dopants may densify the film, limit the number of reactive sites, passivate edge sites, or sacrificially bond with oxygen, all potentially contributing to enhanced oxidation resistance of doped MoS_2 films.

Different mechanisms can improve the coatings' friction, wear, and lifespan. Hard materials densify and harden the film, directly improving wear resistance. On the other hand, soft materials may facilitate the formation of thinner and more uniform MoS_2 films on the coating's surface, indirectly reducing wear. Coatings containing both hard and soft materials, such as $\text{Sb}_2\text{O}_3/\text{Au}$, may benefit from both mechanisms. However, it is essential to note that wear is often inversely related to friction, especially at low temperatures. Table 2 illustrates the development of different doping materials used to enhance the tribological property of MoS_2 .

In the next sections, we will discuss several doping materials that have been used individually. How they have been incorporated into MoS_2 and what type of tribological advancements they achieved will be discussed for individual materials.

4.1 Chromium (Cr)

Chromium (Cr) doping improves MoS_2 's tribological performance, addressing its poor durability in humid and oxidative environments [46, 84, 108, 109]. MoS_2 's hygroscopic nature leads to increased friction, oxidation, and wear, limiting its practical use. Cr-doped MoS_2 coatings, primarily deposited via unbalanced magnetron sputtering and multi-target co-sputtering [46, 108] enable precise control over Cr content. Under humid conditions, Cr forms a protective Cr_2O_3 layer on the coating surface [46], preventing MoO_3 formation and preserving MoS_2 's lubricating properties [84]. Additionally, Cr doping enhances hardness through solid solution hardening, increasing it to 7.5 GPa compared to 0.7 GPa in pure MoS_2 [108]. This denser microstructure improves wear resistance while maintaining a low friction coefficient across various environments. Figure 6 shows how the addition of Chromium can increase the density of the MoS_2 structure.

Cr also stabilizes the tribolayer, forming a lubricating Cr–Mo–S transfer layer on the counter surface, reducing direct contact and ensuring sustained low friction and wear, even in humid conditions [108].

4.2 Silver (Ag)

Silver (Ag) doping enhances MoS_2 's tribological properties by reducing friction and wear due to Ag's low shear strength. It also inhibits MoS_2 oxidation, preserving its performance in atmospheric and high-temperature conditions [34, 84, 110]. Ag doping has been achieved using chemical vapor deposition (CVD) and Magnetron Sputtering. CVD, operating at high temperatures, facilitates Ag dissolution in the MoS_2 matrix, improving mechanical properties and wear resistance [34]. In contrast, magnetron sputtering enables doping at lower temperatures, making it suitable for substrates like p-type Si (100) [84].

At elevated temperatures (~ 350 °C), Ag reacts with MoS_2 oxidation products to form $\text{Ag}_2\text{Mo}_4\text{O}_{13}$, a layered compound with weak Ag–O bonds that shear easily, reducing friction and wear [86]. Ag-doped MoS_2 coatings exhibited a significantly lower wear rate (5.4×10^{-6} mm³/N·m) compared to undoped MoS_2 . Additionally, Ag promotes the formation of a dense transfer layer on friction surfaces, protecting against abrasive wear and improving durability. The coefficient of friction (COF) of Ag-doped MoS_2 dropped to ~ 0.1 within 6 h of sliding, whereas undoped MoS_2 showed a rapid COF increase after only 9 min [34].

Table 2 Popular dopants used to enhance the tribological property of MoS₂

Dopant	Substrate	Deposition Method	Application	Literature
Undoped	100Cr6 bearing steel, AISI 440C steel, AMS 5898 stainless steel, steel AISI 316L + bronze CuAl ₁₀ NiFe ₄ Mn ₁ [91] QT500-7 (Ductile Iron), ZY331608 and ZCuSn ₁₀ Pb ₁₁ Ni ₃ (Bronze Alloy)	Physical vapor deposition (magnetron sputtering), plasma spraying treatment	Bearing application, aerospace, space application, vacuum application, piston pump	[91–95]
Nickel (Ni)	Stainless steel	Sputtering	Space application	[40, 96]
Chromium (Cr)	Si(100)	magnetron sputtering [46, 84, 85]	Vacuum, atmospheric application	[46, 84, 85]
Vanadium (V)	Si (100) wafers, AISI 440C steel	Magnetron sputtering	Vacuum, atmospheric application	[46, 97]
Antimony (Sb)	Titanium nitride	Chemical vapor deposition	High temperature	[34]
Silver (Ag)	Titanium nitride	Chemical Vapor deposition, magnetron sputtering	High temperature, atmospheric application	[34, 84]
g-C ₃ N ₄		Hydrothermal synthesis	Atmospheric application	[56]
Aluminum	Si(100)	Magnetron sputtering	Atmospheric application	[46]
Titanium	Si(100), 304L stainless steel	Magnetron sputtering	Atmospheric application	[43, 46]
Niobium	Si(100)	Magnetron sputtering	Atmospheric application	[46]
Ce–Ti	Si(100)	Magnetron sputtering	Bearing	[98]
Graphene	GCr15 steel	Air spraying	Aviation	[99]
Lead (Pb)	304L stainless steel	Dip spin	Green application	[57]
Sb ₂ O ₃	304L stainless steel	Dip spi	Green application	[57]
Zirconium (Zr)	Si(100) [44], 304L stainless steel	Magnetron sputtering	Humid environment	[44]
Silicon	Si(100), 304L stainless steel	Magnetron sputtering		[43, 45]
TiO ₂	AISI 52100 steel, AZ31B Magnesium Alloy	Drop casting, electrophoretic deposition	Solid lubrication, hydrophobic application	[100, 101]
Copper (Cu)	Si(100)	Magnetron Sputtering	Solid lubrication	[42]
PTFE	Steel	Spray coating	Atmospheric application	[102]
Graphene oxide	M2 steel substrate	Hydrothermal method	Aerospace	[103]
TiCN	304 stainless-steel	Oblique angle Deposition	Aerospace	[104]
Carbon nanotube	Steel	Plasma spraying	Solid state lubrication	[105]
Cr ₃ O ₄	AISI 440C steel wafers, Si(100)	Liquid phase synthesis	Bearing	[106]
Mo ₂ N	Si(100), high speed steel	Magnetron sputtering	Atmospheric application	[107]

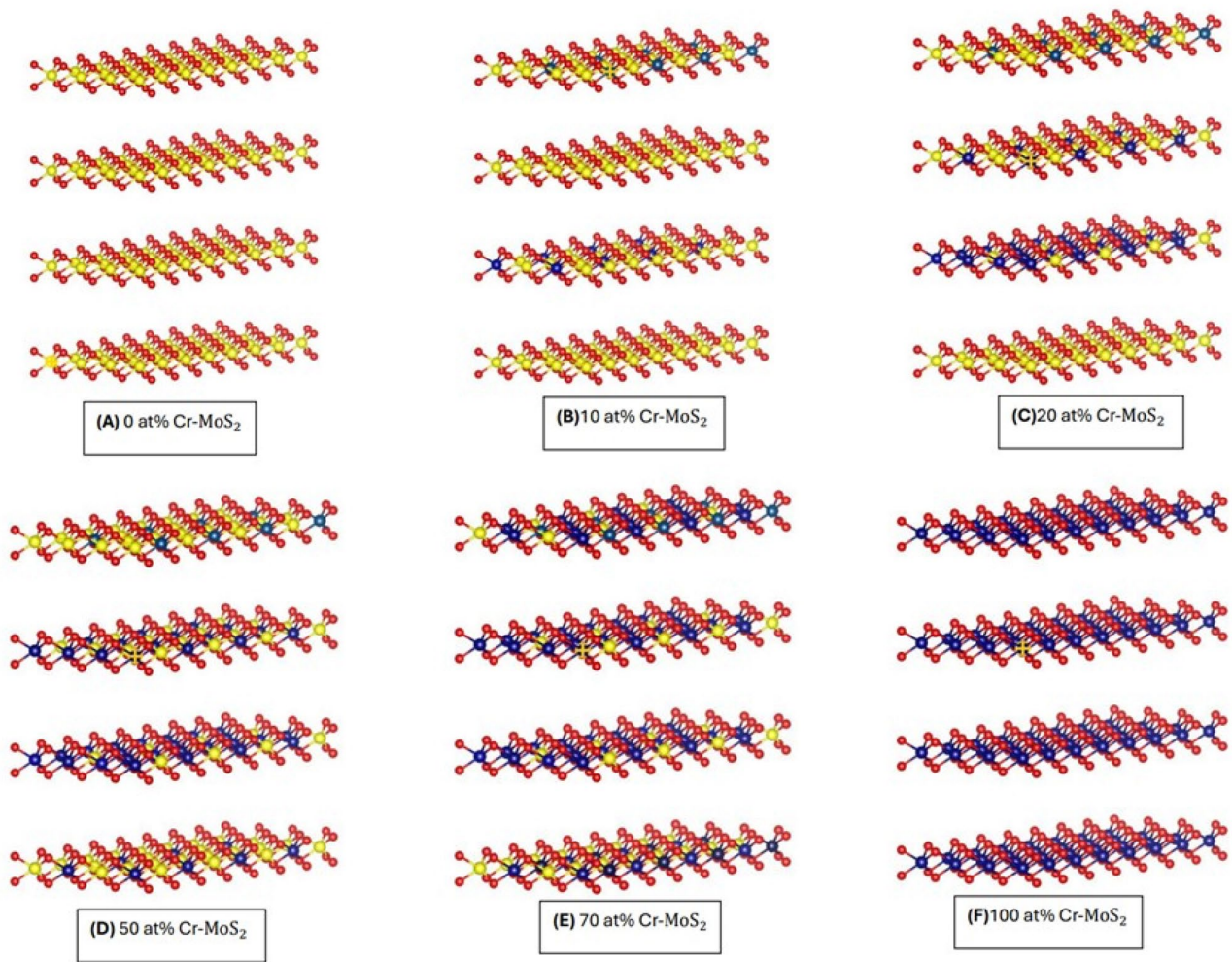


Fig. 6 Atomic structure of MoS₂ for different percentages of chromium doping

4.3 Mo₂N

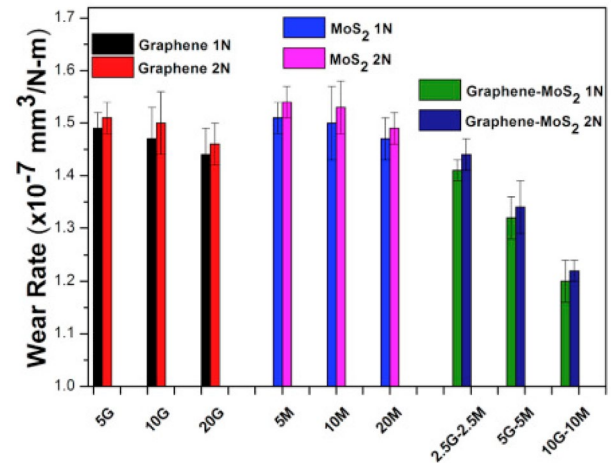
Mo₂N, known for its low friction coefficient and high hardness, enhances the tribological properties of MoS₂ gradient films by improving wear resistance and bonding strength. Its incorporation into MoS₂ films reduces wear in sliding applications while increasing durability under mechanical stress. Combining Mo₂N's hardness with MoS₂'s lubricity optimizes tribological performance, making these composite films highly effective.

However, challenges remain, including the limited number of gradient layers (typically fewer than five), abrupt composition transitions, and unclear effects of layer thickness and composition gradient on overall performance. Future research should focus on optimizing gradient layer thickness, refining composition transition rates, and increasing layer count to develop MoS₂/Mo₂N gradient films with superior industrial tribological properties [111].

4.4 Carbon nanotube (CNT)/carbon

Recently, advancements in the doping process of MoS₂ with carbon nanotubes (CNTs) and other carbon-based materials have shown promising improvements in tribological performance, particularly when measuring the resistance to wear and the reduction of friction [112]. The doping of CNTs with MoS₂ has been accomplished through a variety of fabrication processes such as hydrothermal [113, 114], chemical vapor deposition (CVD) [115], and plasma spraying methods [105], all of which allow MoS₂ to be uniformly dispersed on the surfaces of the CNTs. These fabrication processes are essential

Fig. 7 Wear rate analysis of graphene-MoS₂ solid lubricants. The results show that the composite structure performs better than both materials. (Reused with permission from Elsevier [120])



to utilize the mechanical strength and load-bearing properties of CNTs, adding to the quality of the composite material. It was demonstrated that incorporating CNTs into MoS₂-based composites has decisively reduced the coefficient of friction and the wear rate under varying frictional conditions, mostly due to the formation of a tribofilms that protects the metal surfaces during friction. For example, the inclusion of CNTs into MoS₂ coating thickness has reduced the coefficient of friction from 0.13 to 0.07; it was noted that wt% of 2–4 CNTs which produced optimum results contributing to reductions in friction and improved wear resistance. Furthermore, as the functional groups (oxygen and nitrogen) in the each modified CNT interacted with the sulfur atoms in MoS₂, they reinforced stability of the tribofilm while providing long-lasting protection against abrasive wear. In addition, the introduction of a CNTs also improved mechanical properties of a composite while supporting self-lubricating properties through the environment of nano-bearings between sheets of MoS₂, thereby improving coating durability and efficiency under high-performance tribological applications such as self-lubricating bearings and gas-driven gears [116].

4.5 Graphene

Graphene, due to its *sp*²-hybridized carbon structure, exhibits remarkable strength and chemical inertness, which helps reduce friction and improve wear resistance when used in conjunction with MoS₂. The graphene layers enhance the overall durability of the coating by limiting the penetration of oxygen and water, which are known to accelerate the degradation of MoS₂. Furthermore, graphene's ability to form few-layer structures in tribological contacts improves load-bearing capacity, providing greater wear resistance under higher loads [117, 118].

Graphene can be added to the magnetron-sputtered MoS₂ matrix both before or after the deposition of MoS₂. In [119] graphene is applied via a dripping technique after the deposition of MoS₂. A graphene suspension is prepared and then dripped onto the surface of the MoS₂-based films. This approach ensures even distribution of graphene layers, which improves the wear resistance and frictional properties of the coating.

In combination with laser texturing techniques, graphene-coated MoS₂ surfaces exhibit significantly reduced friction, with coefficients as low as 0.2, and improved wear lifetimes. The texturing process creates grooves that act as reservoirs for the graphene lubricant, allowing for a sustained release of the material during sliding, further optimizing the tribological performance [118]. The addition of graphene stabilizes the MoS₂ structure, preventing oxidation and maintaining a low friction coefficient, even at elevated temperatures [117, 119]. Figure 7 shows how graphene has improved the wear properties of MoS₂.

4.6 Graphene oxide (GO)

Graphene oxide (GO) enhances the tribological properties of MoS₂ by improving its wear resistance, load-bearing capacity, and oxidation protection. MoS₂'s lamellar structure provides low friction, but its performance degrades under humid conditions, mechanical strain, and oxidation [121, 122]. GO, with its high mechanical strength, large surface area, and functional groups, synergistically interacts with MoS₂ to minimize friction and wear [103, 114].

GO/MoS₂ composites enhance lubrication through the ease of MoS₂ layer shearing, GO's protective structure against moisture absorption, and the formation of a transfer film reducing wear and friction [123]. Hydrothermal synthesis of RGO/MoS₂ composites ensures better dispersion of MoS₂ nanosheets within GO layers, creating a stable lubricating layer [121]. Electrophoretic deposition (EPD) on metallic substrates, such as stainless steel, has shown lower COF, superior wear resistance, and oxidation protection [120].

GO/MoS₂ films exhibit a 50% reduction in friction coefficient compared to pristine GO films and extended durability over pure MoS₂. Under high load conditions, MoS₂'s COF dropped from 0.14 at 5 N to 0.03 at 100 N, demonstrating its enhanced tribological performance [103].

4.7 Polytetrafluoroethylene (PTFE)

PTFE is a polymer characterized by its superior chemical inertness, low surface energy, and water resistance [124], making it a very effective additive for enhancing the durability and performance of MoS₂-based lubricants. Several fabrication and synthesis methods exist to combine PTFE with MoS₂ effectively. The most well-known method is co-sputtering, which deposits PTFE and MoS₂ simultaneously onto substrates by a combination of direct current (DC) and radiofrequency (RF) magnetron sputtering [125]. The co-sputtering method allows for precise control over the content of PTFE in the film by shifting the sputtering power and co-deposition time. PTFE was also added into MoS₂ by hot press molding techniques, which involves pressing the mixture together at elevated temperatures and high pressures for uniform dispersion, and bonding strength. Another way PTFE has been combined into MoS₂ is with the embedding of inorganic matrices, such as phosphate coatings to give the coating mechanical flexibility and enhance the tribological properties of the coating under extreme operating conditions [126].

PTFE significantly improves key tribological properties. The polymeric nature of PTFE allows for the characteristics of improved film hardness and modulus of elasticity, but most importantly, PTFE enhanced the reducing of the friction coefficient and wear dirt and does use PTFE is a very non-polar and inert nature which helps not requires high humidity and oxygen, both known to degrade the lubricants performance with the use of MoS₂. The embedded PTFE particularly benefits the enhanced coating ability to build a continuous transfer film in the sliding interface, or surface, which serves as a self-lubricating layer to maintain separation between sliding surfaces, hence, reducing the coefficient of friction (COF) and improving wear resistance. The embedded PTFE retains the integrity of the MoS₂ films after prolonged exposure to high humidity open-air environments, ultimately extending the life for a working lubricant [125].

The tribological properties of PTFE are of significant value to the improvement of MoS₂'s tribological behavior. As a high-molecular-weight long-chain polymer, PTFE can fill in micro-gaps and surface topographies to produce a smoother sliding surface. In addition, due to low surface energy, PTFE will assist in the formation of a stable low shear surface, which is important for reducing friction [102]. In composite formulations such as PI-MoS₂/PTFE, PTFE also assists in the formation of transition layers and oriented shear layers that reform through the run-in period to lower the COF, particularly under vacuum when MoS₂ alone may not provide sufficient lubricant fluid [126].

PTFE doping of MoS₂ has seen significant advances over the years for the purpose of enhancing tribological properties of MoS₂. While limited, testing in laboratory environments has shown low COF, enhanced hardness and wear resistance for PTFE-doped MoS₂ films, relative to pure MoS₂ films. For instance, MoS₂-PTFE co-sputtered thin films showed outstanding to nearly perfect performance after long-term storage in open-air environments, maintaining COF under 0.2 through various sliding tests. Certain extreme conditions, such as vacuum or atomic oxygen (AO) irradiations, confirmed the beneficial frictional attributes of PTFE-MoS₂ composites and additives [127].

4.8 Copper (Cu)

The utilization of pure molybdenum disulfide (MoS₂) as a solid lubricant has limitations based on its relatively low hardness and high coefficient of friction, especially under atmospheric conditions. In order to overcome this limitation, copper (Cu) has been incorporated into MoS₂. Copper's metal-like properties allow for the formation of a turbulence-like micro-nano structure (TL-MoS₂) inside the MoS₂ matrix, which significantly increases hardness and tribological performance [42, 128].

A variety of fabrication and synthesis methods have been developed to produce effectively Cu-doped MoS₂. Magnetron sputtering, for example, is bimodal which allows for control of Cu/MoS₂ films with respect to their desired microstructure [42]. Self-propagating high-temperature synthesis (SHS) or in-situ reduction also produced nanolamellar MoS₂ doped

with Cu nanoparticles throughout the synthesis, thus ensuring a uniform distribution of Cu [129]. Effective tribological performance demands the well-dispersed Cu in the MoS₂ matrix.

Doping Cu into the MoS₂ structure results in a change in its crystal orientation from (100) in pure MoS₂ to (002) in Cu/MoS₂. This reorientation of the crystal structure and densification of the film results in a hardness ten times that of pure MoS₂. The turbulence-like structure will extend the strength of the planar MoS₂ network into a three-dimensional form, increasing wear resistance and oxidation resistance [42].

On a tribological performance level, Cu-dopants exhibit a significantly lower friction coefficient, reduced to 0.07 in ambient atmosphere, which is approximately three times lower than pure MoS₂ [130]. This decrease in friction coefficient is a result of the MoS₂ tribofilm generated during sliding wear with Cu acting as a catalyst. Cu and MoS₂ produced a compelling effect during sliding wear, especially in producing a stable transfer film which represents excellent lubrication performance thus making Cu-doped MoS₂ viable lubrication candidates.

4.9 Titanium oxide (TiO₂)

Adding TiO₂ into MoS₂ has proven to give an extremely improved tribological property to MoS₂ [131]. Pure MoS₂, despite its popularity, is used as a solid lubricant; however, it has poor wear resistance. In general, the incorporation of nanoparticles of TiO₂ is made within the matrix of MoS₂ during the synthesis process, indeed improving the overall performance. Such an addition of 15–20 wt% TiO₂ nanoparticles, along with proper pre-treatments aimed at increasing substrate porosity, for instance, by phosphating, enhances the adhesion at the coating-substrate interface [132]. Therefore, there will be improved hardness, good wear resistance, and friction reduction. In one test, the addition of 27.69 nm crystallite size TiO₂ nanoparticles to a MoS₂ matrix reduced the coefficient of friction by 31% and wear rate by 39% compared to the pure MoS₂ coating [133]. This results from a stable lubricating film that might be formed with support from the interaction between MoS₂ and TiO₂. TiO₂ strengthens the structure of the coating, hence increasing its endurance under high-contact pressures and sliding conditions. However, there is one critical point: above 15 wt% of TiO₂ [100], the coatings become brittle, showing a clear reduction in scratch resistance. Thus, an ideal MoS₂-TiO₂ composite commonly possesses a balanced composition of typically 60% MoS₂ and 20% TiO₂, balancing the friction-reducing potential with maintaining durability [134]. Therefore, such improvements make the MoS₂-TiO₂ composites rather effective in industrial applications requiring advanced lubrication and wear resistance [135].

4.10 Zirconium (Zr)

Zirconia has been used as a dopant in molybdenum disulfide solid lubricants in the last few decades, mainly for environmental applications involving high humidity and corrosive media [136]. Till now, Zr was generally added to MoS₂ coatings by magnetron sputtering and co-sputtering or in microwave-assisted synthesis, where either Zr was co-deposited with MoS₂ or the Zr nanoparticles were introduced into the MoS₂ matrix [44, 137, 138]. In particular, Zr addition in its tetragonal phase alters the microstructure of MoS₂ from crystalline to amorphous with dispersed Zr nanocrystallites [138]. This transformation leads to a denser, compact coating with reduced roughness and enhanced hardness. These significantly improve wear resistance and adhesion through increased hardness and reduced grain size, hence giving lower CoF and wear rates under various environmental conditions [44, 138]. Further, the Zr forms a stable ZrO₂ layer on the surface that protects against oxidation and corrosion in a humid environment. It prevents the oxidation of MoS₂ to MoO₃, which otherwise is a cause for degradation in its lubrication properties [137, 138].

The improvement in the tribological properties of MoS₂-Zr composites may be attributed to several factors. Increased hardness and refinement of grain size increase the load-carrying capacity and lower the wear by reducing the amount of material lost during sliding contact [137]. Besides, the presence of ZrO₂ as a stable oxide layer suppresses galvanic corrosion and enhances the overall durability of coatings [44]. The non-covalent interactions between the Zr nanoparticles and the MoS₂ matrix ensure homogeneous dispersion with better mechanical stability and tribological efficiency upon testing [139]. Adding Zr strengthened the mechanical and tribological properties of MoS₂ while prolonging its lifetime under severe environments; therefore, it has been considered a prospective material for advanced solid lubricants in aerospace and industry.

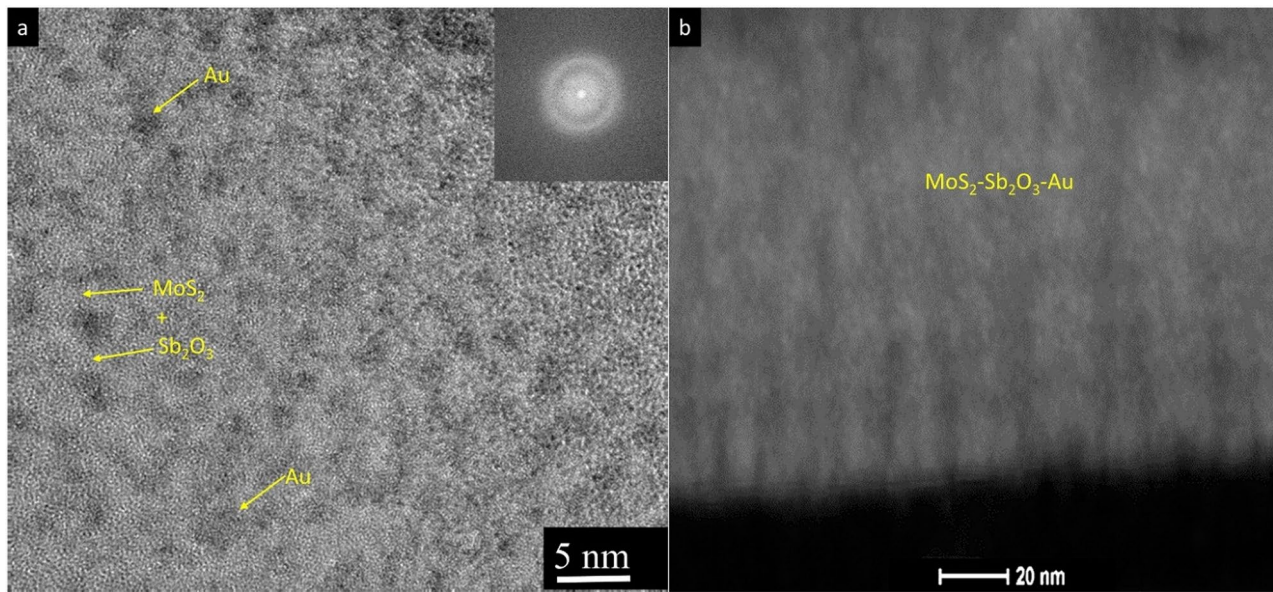


Fig. 8 **a** High-Resolution Transmission Electron Microscopy (HRTEM) and **b** High-angle Annular Dark-field Scanning Transmission Electron Microscopy (HAADF-STEM) image of Sb_2O_3 and Gold doped MoS_2 coating which shows the presence of different atoms in the MoS_2 structure. (Reused with permission from Elsevier [87])

4.11 Antimony oxide (Sb_2O_3)

Adding Sb_2O_3 into MoS_2 solid lubricant significantly improves tribological properties for extremely harsh environments, including high temperature, high humidity, and space environments. Additions of Sb_2O_3 were made into MoS_2 primarily by techniques like magnetron sputtering, burnishing, and rf sputtering, along with other elements like Au or nanodiamond particles [87, 140–143]. Such improvement was attributed to the densification of the structure of MoS_2 by Sb_2O_3 , which reduces the intrinsic porosity in this structure and changes the morphology to a compact, amorphous-like form. Figure 8 shows a TEM image of Sb_2O_3 doped structure and how it has densified the MoS_2 . Under close inspection the difference in compactness can be seen for just gold and $\text{MoS}_2 + \text{Sb}_2\text{O}_3$. The increase in hardness by such densification enhances the wear resistance of the coatings under high contact stress conditions [87, 143, 144]. Additionally, Sb_2O_3 acts as an oxygen scavenger to prevent the oxidation of MoS_2 to MoO_3 , a process that generally destroys its lubricating properties in a humid or oxidative environment [140, 143]. This is particularly effective under low-earth-orbit conditions where atomic oxygen seriously degrades traditional lubricants. It has also been reported that the addition of Sb_2O_3 has appropriately stabilized the MoS_2 structure at high temperatures, allowed for superlubricity with a CoF below 0.01, and enhanced wear lives of coatings [140]. Unfortunately, Sb_2O_3 has proved somewhat effective in providing low friction and high wear resistance of MoS_2 coatings intended for aerospace engineering, even in vacuum and space conditions, by forming a stable, protective layer against environmental degradation. These properties make Sb_2O_3 -doped MoS_2 solid lubricants highly suitable for use in demanding mechanical systems, such as spacecraft components, high-temperature bearings, and other critical moving parts that require reliable lubrication under extreme conditions [87, 141, 143].

4.12 Nickel (Ni)

The addition of nickel (Ni) in molybdenum disulfide (MoS_2) significantly enhances its tribological properties by mitigating oxidation, increasing hardness, and reducing the friction coefficient [145]. MoS_2 's lamellar crystalline structure provides low shear strength between its layers, resulting in excellent lubricating properties. However, under real-world conditions, MoS_2 oxidizes into molybdenum trioxide (MoO_3) when exposed to oxygen and water vapor, increasing friction and wear. Doping with Ni reduces this oxidation while introducing mechanical improvements like increased hardness and superior wear resistance [40, 146].

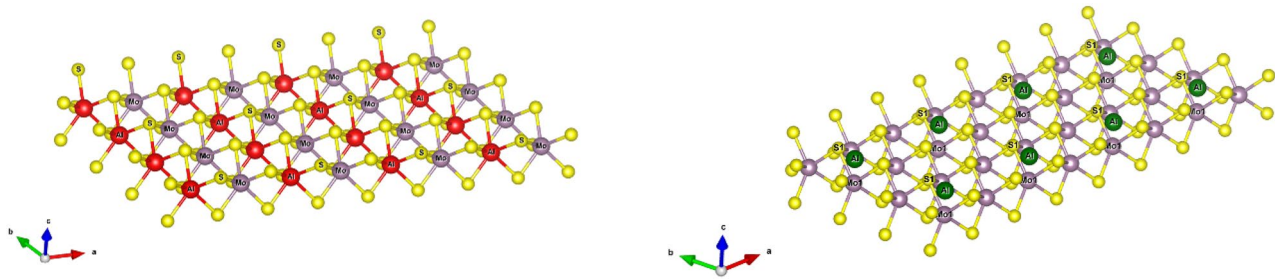


Fig. 9 Structure of Aluminum doped MoS₂

Several fabrication methods are used to produce Ni-doped MoS₂ coatings, each contributing uniquely to its improved performance. Magnetron sputtering is a preferred technique for achieving a homogenous distribution of Ni within the MoS₂ matrix, with Ni content optimized between 5 and 7% to avoid brittleness and maximize tribological benefits [40]. Electroless plating is another widely applied method, enabling the co-deposition of Ni and MoS₂ onto substrates, followed by heat treatment to enhance adhesion and hardness. It is particularly effective for industrial-scale applications, with heat treatments at 300–400 °C improving wear resistance and reducing the friction coefficient from 0.15 to 0.08, even at 500 °C [147]. Laser surface texturing (LST), combined with electrochemical deposition (ECD), further enhances frictional properties by creating microgroove arrays, while electrospark deposition (ESD) ensures metallurgical bonding between Ni-MoS₂ and substrates, resulting in highly wear-resistant coatings [146, 148].

Nickel's primary role as a dopant is to protect MoS₂ from oxidation by acting as a barrier, inhibiting the formation of MoO₃. This protection ensures that Ni-doped MoS₂ maintains low friction even in the presence of oxygen or water vapor, making it suitable for high-temperature applications. Additionally, Ni increases the hardness of the composite, with Ni-MoS₂ coatings showing hardness values around 12 GPa compared to the 1–2 GPa of undoped MoS₂. This enhanced hardness allows Ni-doped MoS₂ to perform well under high-load conditions, extending its service life and durability [40, 146, 147].

The friction coefficient of Ni-doped MoS₂ is also dramatically reduced. Pure MoS₂ exhibits friction coefficients of 0.15–0.25 in humid or oxidative environments, but Ni-doped MoS₂ has achieved values as low as 0.07, representing a 70% reduction. This improvement is attributed to enhanced sliding of MoS₂ layers and the formation of a stable tribofilm facilitated by Ni, which reduces both friction and wear [40, 148].

4.13 Aluminum (Al)

Al has been shown to act as an effective dopant for MoS₂-based coatings in mitigating several problems related to the material's tribological performance, especially under humid and high-wear conditions. As mentioned earlier, MoS₂ has been generally known to be an excellent solid lubricant due to its lamellar structure; however, under oxidative and humid conditions, it is highly degraded and leads to increased friction and wear. The addition of Al to MoS₂ coatings has been found to increase wear resistance, enhanced oxidation resistance, and maintained a friction coefficient in changing environmental conditions, making a very good modification [48, 149].

Al-doped MoS₂ coatings are prepared by different advanced fabrication methods. Among them, magnetron sputtering has been one of the most popular techniques due to the precision it allows in film composition and microstructural control. Al-MoS₂ coatings were prepared by both DC and RF magnetron sputtering techniques, which allow co-deposition onto substrates such as silicon wafers and stainless steel [6, 48]. Besides, PLD has been used to deposit Al-MoS₂ films that can have the advantages of MoS₂ basal plane orientation and enhancement in adhesion by a simultaneous magnetron sputtering of Al [149]. Figure 9 shows the structure of Aluminum doped MoS₂. It is possible, by using PLD along with magnetron sputtering, to precisely control the content of Al and to determine the coatings with an improved wear resistance, even under a humid atmosphere. Different annealing treatments in the temperature range 300–500 °C under inert atmospheres had been conducted for the release of residual stresses and improvement in performance of coatings [6, 149].

Introduction of Al substantially improves the key tribological properties of MoS₂. First, the friction coefficient in Al-doped MoS₂-based coatings is reduced, especially under high humidity, where pure MoS₂ suffers critical losses in

performance. Due to its ability to form oxides, for example, Al_2O_3 , under tribostress, Al builds up a lubricating film that stabilizes the friction coefficient. For example, Al-MoS₂ films deposited by PLD exhibited a friction coefficient as low as 0.08 at 25% relative humidity, which was significantly better compared with the pure MoS₂ films under similar conditions.

Regarding wear resistance, the doping of Al significantly improves it. The addition of Al into the MoS₂ matrix not only provides increased hardness but also acts to improve toughness of the coatings. Consequently, they show better resistance against material loss during sliding wear. Such is the case with the addition of Al in Cu-Al/MoS₂ coatings, improving toughness and thereby reducing wear, as evidenced by the formation of the protective $\gamma_2\text{-Cu}_9\text{Al}_4$ phase [6]. Furthermore, XRD studies have demonstrated that Al-alloyed MoS₂ coatings tend to adopt multilayer structures which contribute to improved endurance and stability during wear [150].

The major mechanism for improved performance of Al-doped MoS₂ coatings is through the interaction of Al with the sliding MoS₂ matrix. Hence, Al acts as an oxygen scavenger to inhibit the oxidation of the MoS₂ into MoO₃, which is deleterious to lubrication. The alumina formed during wear, Al_2O_3 , develops an adhesive layer with low friction and prolongs wear life in most coatings having high humidity [149, 150]. The interaction prevents the breakdown of the MoS₂ lamellar structure, so its tribological properties are stabilized for a long time.

4.14 Titanium (Ti)

Ti doping effectively reduces the sensitivity of MoS₂ to oxidation by forming a protective layer of titanium oxides. In [151], when the Ti content was optimized at 13.48 at%, the oxidation resistance of the Ti-MoS₂ films significantly enhanced, as this composition prevented MoS₂ from reacting with environmental moisture and oxygen. This protective effect allows the material to maintain low friction and stable wear performance, even in harsh environments, extending the operational lifespan of the coatings.

In addition to oxidation resistance, Ti doping improves the mechanical properties of MoS₂, particularly its hardness. Pure MoS₂ is known for its relatively low hardness, which makes it prone to wear under high-load conditions. By incorporating Ti, the hardness of the MoS₂ coatings increases substantially. In one study, Ti-doped MoS₂ coatings produced via mechanical ball milling achieved a hardness of 831.9 HV, representing a 28.93% increase compared to untreated samples [152]. This significant improvement in hardness leads to better wear resistance, with Ti-doped samples showing a 51.23% reduction in wear volume under a load of 80 N [152, 153]. These improvements are due to Ti's enhanced structural integrity, which strengthens the MoS₂ matrix and reduces the rate of material degradation during sliding.

The frictional properties of Ti-MoS₂ are also markedly improved. Ti-doped MoS₂ coatings can achieve ultra-low friction coefficients, particularly in vacuum environments. Ti-MoS₂/WC multilayer films achieved a friction coefficient as low as 0.009 in a vacuum [154]. Even after exposure to moisture and salt spray environments, these films maintained excellent frictional performance, indicating that Ti doping not only improves wear resistance but also enhances the environmental adaptability of MoS₂ [154] (Fig. 10).

Several fabrication methods have been employed to synthesize Ti-doped MoS₂ coatings, each contributing to the improved performance. Magnetron sputtering is a commonly used technique, offering precise control over the Ti content and enabling the production of uniform, dense films. For example, magnetron-sputtered Ti-MoS₂ films with Ti contents optimized between 11.13 and 17.90% exhibited enhanced hardness and wear resistance due to better crystallization and densification [86]. Mechanical ball milling has also been used to create Ti-MoS₂ coatings with nano gradient structures, which contribute to both surface hardness and wear resistance. In one study, ball-milled Ti-MoS₂ coatings exhibited a 28.93% increase in hardness and a significant reduction in wear volume [152]. Ultrasonic impact treatment (UIT) has been employed to simultaneously fabricate surface Ti-MoS₂ coatings and gradient structures, reducing wear volume by up to 89.78% under certain loads [153]. Figure 11 shows the microstructure of Ti doped MoS₂.

However, excessive Ti content can lead to increased surface roughness, as well as the formation of brittle phases, which can negatively affect the coating's overall performance. Films with higher Ti content showed increased friction due to abrasive wear in certain conditions, underscoring the importance of optimizing the Ti content for specific applications [152].

4.15 Lead (Pb)

The structure of MoS₂ is susceptible to environmental factors. By incorporating Pb into MoS₂ coatings, researchers have observed that Pb promotes a more compact structure, helping to prevent oxygen and water from permeating the film

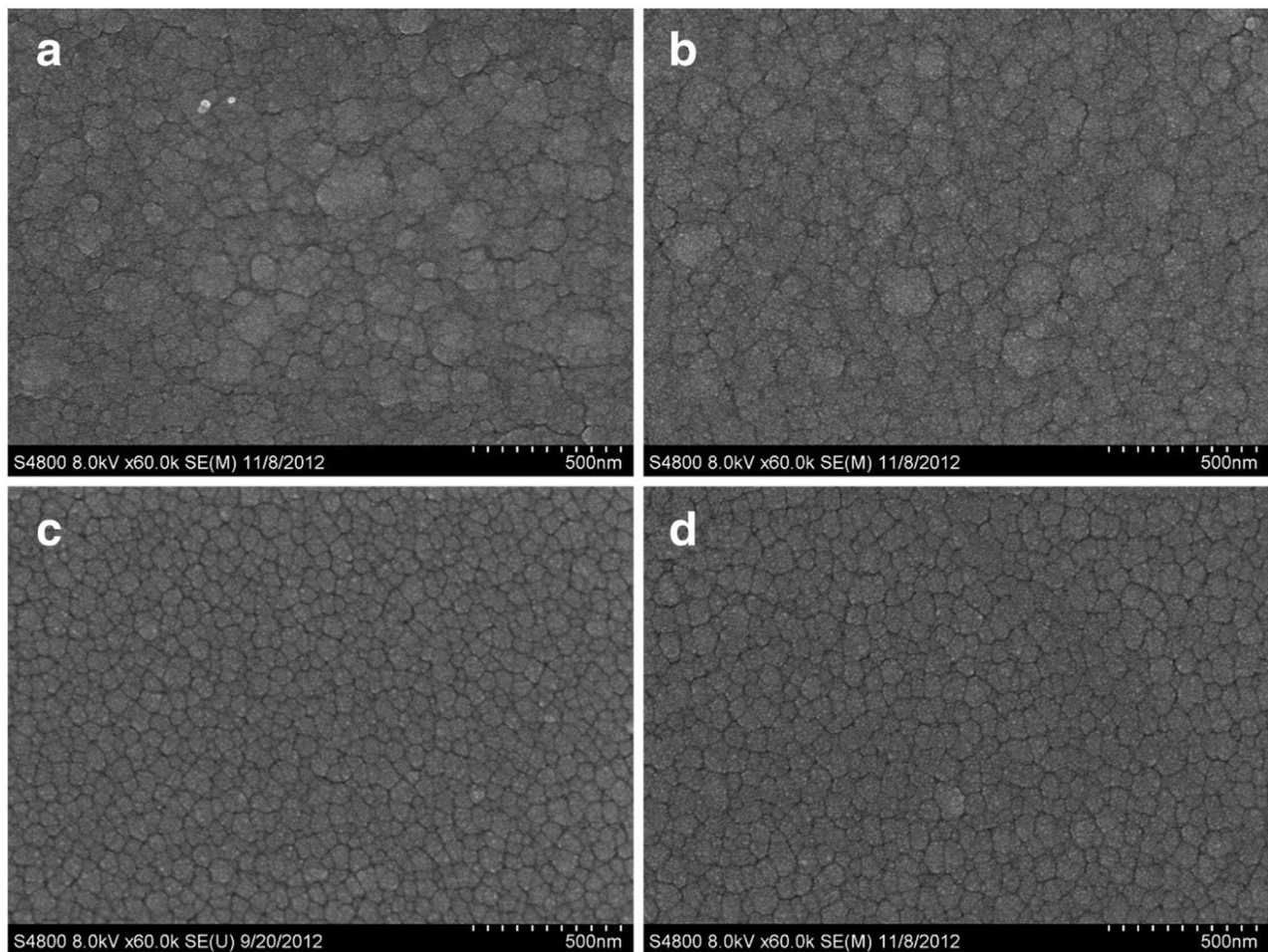


Fig. 10 SEM images of different MoS₂-Ti composite coatings with different percentages of Titanium. **a** 0%, **b** 4.6%, **c** 13.5% and **d** 19.9%. The image illustrates how the infusion of titanium has modified the microstructure of MoS₂ which eventually results in different tribological properties. (Reused with permission from Elsevier [155])

[156]. Additionally, Pb's ability to preferentially oxidize to PbO provides a protective barrier that inhibits further oxidation of [157] MoS₂, enhancing the longevity and performance of the coating [157–159].

By doping MoS₂ with Pb, several mechanisms come into play to mitigate these issues. Pb enhances the compactness of the MoS₂ film, leading to a reduction in porosity, which is critical for minimizing the infiltration of oxidizing agents like water and oxygen. This densification effect strengthens the overall structure, making it less prone to chemical degradation in harsh conditions. Pb also forms a passive oxide layer (PbO) during exposure to oxygen, which acts as a sacrificial barrier, preventing further oxidation of MoS₂. This self-limiting oxidation mechanism helps preserve the integrity of the MoS₂ layer, allowing it to maintain its lubrication properties over longer periods [157, 160].

In addition, Pb has a low shear strength, which contributes to reducing the overall friction of the MoS₂ coating. When doped with Pb, the MoS₂ layers exhibit better basal plane orientation, meaning the planes responsible for low-friction sliding are aligned parallel to the substrate. This alignment promotes easier shear and lowers the friction coefficient, even under vacuum and humid conditions. The structural refinement brought about by Pb doping not only enhances frictional properties but also improves the mechanical robustness of the coating, resulting in higher hardness values and better wear resistance [160].

Lead (Pb) doping in MoS₂ coatings, especially in a multilayered structure, significantly enhances tribological performance in high-temperature conditions. The alternating layers of Pb and MoS₂ create a denser film, improving the basal-plane orientation of MoS₂ and reducing friction (below 0.1) [119].

The deposition of Pb-doped MoS₂ films is typically achieved through magnetron sputtering, which allows for precise control over the film composition and structure. The hardness of these films has been shown to increase significantly due

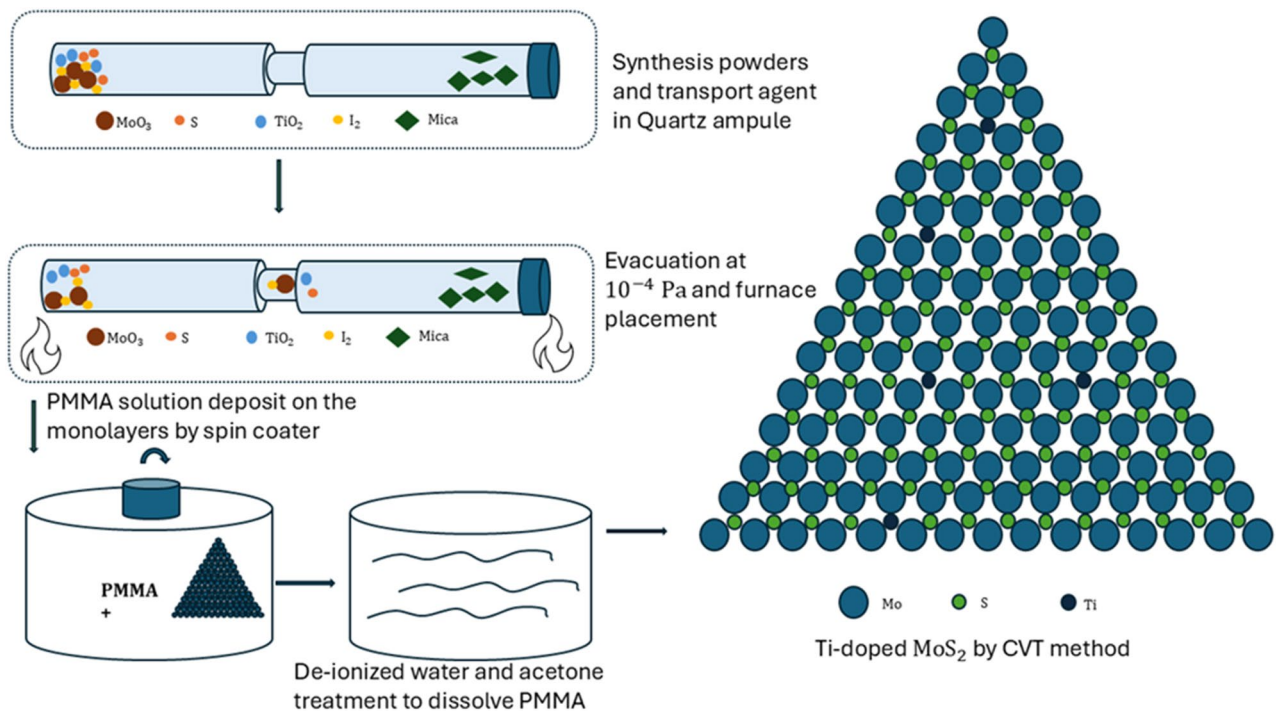


Fig. 11 Ti-doped MoS₂ monolayers synthesized using the chemical vapor transport (CVT) method. A mixture of MoO₃ (99.7%, ~2.30 mg), TiO₂ (99.99%, ~0.40 mg), Sulfur (99.7%, ~0.40 mg), and Iodine (99.7%, ~3.00 mg) as a transport agent was prepared and loaded into a vacuum-sealed quartz ampoule ($1\text{--}5 \times 10^{-4}$ Pa) to prevent iodine evaporation

to the denser microstructure, with values ranging from 5.9 to 7.2 GPa. The friction coefficient of these films in vacuum remains below 0.25, and the wear rate reaches a remarkably low value of 1.14×10^{-8} mm³/N·m, indicating a major improvement in durability and performance over undoped MoS₂ [157, 158, 160].

4.16 Vanadium (V)

Pure MoS₂ has excellent lubricating properties in vacuum, but its porous structure makes it susceptible to oxidation and erosion when exposed to AO. This results in a rapid degradation of its tribological performance. The key role of V doping is to mitigate these issues by enhancing the film's structural integrity and oxidation resistance [161, 162].

Vanadium plays a critical role in modifying the microstructure of MoS₂. By introducing V, the coating transitions from a loose, columnar crystal structure to a denser, more compact form. This reduction in porosity is essential because it limits the access of oxygen and moisture, which are responsible for the oxidative degradation of MoS₂. V also forms a protective vanadium oxide (V₂O₅) layer during AO exposure, which acts as a barrier against further oxidation of MoS₂, maintaining the coating's integrity under harsh conditions [161].

Additionally, V doping contributes to enhanced mechanical properties. The denser structure reduces wear and improves hardness, allowing the coating to maintain a low friction coefficient, even under high stress or in vacuum environments. V-doped MoS₂ films exhibit a friction coefficient as low as 0.04 and a significantly lower wear rate of 1.29×10^{-6} mm³/N·m, compared to undoped MoS₂. This makes V a key dopant for improving the longevity and durability of MoS₂ coatings, especially in space applications where exposure to AO is inevitable.

The controlled incorporation of V is typically achieved through magnetron sputtering, followed by thermal sulfurization. This process ensures uniform V distribution within the MoS₂ matrix, enhancing both the structural and tribological properties of the coating [162]. V is thus used to overcome the primary limitations of MoS₂ oxidation and wear making it an essential element for improving the performance of MoS₂ in extreme conditions.

The addition of different dopants has different advantages and disadvantages. Different dopants are used for different applications also. Table 3 gives an overview of the impact of different dopants on the tribological properties of MoS₂.

Table 3 Overview of dopant properties

Doping material	Advantages	Disadvantages	Applications	References
Nickel (Ni)	Enhances oxidation resistance, increases hardness, reduces friction coefficient	It can cause brittleness at high concentrations, and requires precise deposition control	Space applications, high-temperature lubrication, aerospace	[42, 52]
Chromium (Cr)	Forms Cr ₂ O ₃ protective layer, improves wear resistance, enhances hardness	This can lead to excessive hardness, reducing the flexibility of the coating	Vacuum, atmospheric applications, high-temperature coatings	[47]
Vanadium (V)	Enhances oxidation resistance, improves mechanical properties, reduces wear	High processing complexity, can introduce brittleness	Vacuum, high-stress environments, aerospace applications	[48]
Antimony (Sb)	Densifies structure, reduces porosity, acts as an oxygen scavenger	Excessive amounts increase brittleness, limiting wear resistance	High-temperature coatings, aerospace applications	[34]
Silver (Ag)	Reduces friction, enhances load-bearing, forms stable transfer layer	Reacts at high temperatures, leading to possible wear degradation	High-temperature environments, industrial lubrication	[63]
Aluminum (Al)	Improves oxidation resistance, and reduces friction in humid conditions	It may increase surface roughness and requires precise composition control	Aerospace, mechanical components, industrial coatings	[59]
Titanium (Ti)	Enhances hardness, reduces oxidation sensitivity, improves wear resistance	Excessive Ti can increase roughness and reduce overall performance	Aerospace, mechanical wear coatings, vacuum lubrication	[47, 52]
Niobium (Nb)	Enhances mechanical strength, improves oxidation resistance	High cost of deposition, potential for brittle phases	High-performance wear coatings, aerospace applications	[46]
Zirconium (Zr)	Forms stable ZrO ₂ layer, improves wear resistance, enhances durability	Can become brittle at high concentrations, reducing flexibility	Corrosive environments, high-humidity applications	[44]
Silicon (Si)	Reduces oxidation, improves adhesion, enhances wear resistance	High doping levels can degrade lubricity	Tribological coatings, aerospace applications	[45]
Titanium oxide (TiO ₂)	Enhances hardness, reduces friction, improves adhesion	Excessive TiO ₂ can cause brittleness, reducing scratch resistance	Industrial coatings, wear-resistant applications	[38, 49]
Copper (Cu)	Forms turbulence-like microstructure, enhances hardness, reduces friction	Too much Cu can alter MoS ₂ structure unfavorably	Solid lubrication, industrial wear applications	[42]
Graphene	Improves wear resistance, reduces oxidation, stabilizes MoS ₂ structure	Graphene layer stacking can reduce lubricity under high loads	Aerospace, vacuum lubrication, high-stress environments	[37]
Lead (Pb)	Enhances structural integrity, reduces porosity, forms protective PbO layer	Pb toxicity is a concern, limiting applications	High-temperature wear resistance, aerospace, industrial lubrication	[9]
Carbon nanotube (CNT)	Improves lubrication, reduces wear, enhances durability	Difficult to uniformly disperse, requires precise synthesis methods	Aerospace, self-lubricating bearings, industrial applications	[55, 70]

5 Mechanism behind low friction and wear of doped MoS₂

The mechanisms through which dopants enhance the tribological and oxidation properties of MoS₂ are a topic of active research and a crucial aspect of materials science and engineering. In terms of oxidation resistance, several mechanisms have been proposed. Dopants are believed to influence the growth of MoS₂ crystallites, reducing their size and increasing density [81], which can improve oxidation resistance by reducing crystalline breakage [89] and the number of available edge sites for oxidation [19]. Dopants may also passivate edge sites, making them less reactive with oxygen. Furthermore, dopants existing as a separate phase could bond with oxygen to form an oxide [149].

Various mechanisms have been used to explain the effect of dopants on the wear and friction properties of MoS₂, with significant implications for practical applications. The most commonly cited mechanism is the increase in hardness, which results from the distortion of the MoS₂ crystal structure in doped films [90, 138, 163, 164]. This increased hardness directly benefits wear, a crucial factor for the coating's lifespan in applications such as space missions, where coatings must endure for many years without replacement. Some studies report that dopants can also increase the elastic modulus of the layer. Hardness is associated with increased density and improved friction and wear [165, 166]. However, increased hardness is not the only mechanism, as some dopants decrease hardness yet improve performance. The suggested mechanism is that the dopant acts as a separate nanoscale phase, providing beneficial properties that affect the transfer film [166, 167].

Various factors influence the specific mechanisms by which a dopant improves coating performance. These include the dopant type, concentration, and conditions during coating deposition and operation. In the following sections, we will discuss the effects of some commonly studied dopants, systematically grouped by proposed mechanism.

5.1 Improving tribological properties through hardening

Ti is among the most extensively studied dopants for improving tribological performance through hardening [108, 168]. It is suggested that Ti atoms either intercalate between MoS₂ layers or substitute for Mo, as evidenced by the lack of change in the crystal c-axis after doping [7, 169]. Ti atoms are evenly distributed in the MoS₂ matrix without forming Ti-rich clusters or precipitates [170]. Coatings with 16% Ti were slightly sub-stoichiometric in sulfur and amorphous. These Ti-doped coatings exhibit a columnar platelet structure, with higher Ti concentrations leading to increased density. Ti enhances coating hardness, but additional Ti has a detrimental effect beyond a specific concentration [171]. Tribological properties also improve with increasing Ti content to an optimal value of 18% [169]. Ti doping improves oxidation resistance, as shown by consistent friction measurements with 10.8% Ti across various humidities [171]. The performance of Ti-doped MoS₂ depends on the counterface material, with ZrO₂ performing best among the brass, GCr15 steel, WC, and ZrO₂ [164].

Similarly, the hardening mechanism has been used to explain the beneficial effects of Cr, Zn, and Ni dopants in improving friction and wear. The optimum concentration of Cr for hardness was 16.6%, while for friction and wear, it was 10% [108]. Zr-doped coatings consist of nanocrystalline Zr in an amorphous matrix, with optimal performance observed at a 10% concentration [138]. Ni has shown promise as a dopant, particularly in mitigating the increase in friction at low temperatures commonly observed with MoS₂ and other doped MoS₂ coatings [81]. Ni dopants are used in various space components due to their cost-effectiveness and low-temperature performance. Metal dopants generally enhance performance up to a maximum concentration, beyond which layering occurs, leading to poor tribological performance [88]. Lastly, doping with Sb₂O₃ results in a fine-grained microstructure and a two-phase system with MoS₂ crystallites dispersed in amorphous antimony oxide [89]. Sb₂O₃-doped coatings exhibit increased density and hardness, with an optimal concentration of 35% [89].

5.2 Improving tribological properties through transfer film

The most extensively studied dopant that does not increase hardness is Au [19, 163, 165–167, 172]. In Au-doped MoS₂, nanocrystalline Au particles are dispersed in the MoS₂ matrix, with the particle size dependent on temperature [172]. Unlike Ti, Au doping decreases hardness [19], so tribological properties are improved through a different mechanism. It is proposed that subsurface coarsening of Au nanoparticles provides load support, allowing shear of surface MoS₂ basal planes [166].

A direct comparison of MoS₂, Au, and Au-doped MoS₂ (at relatively high concentrations > 42%) showed that the doped coating was superior to MoS₂ and Au alone [167, 172]. Moreover, the optimal amount of Au depended on

contact stress, with less Au being better at high stress and more Au being better at low stress. This observation led to the hypothesis that Au provided the optimal amount of MoS₂ in contact at high stress. In contrast, more Au limited the amount/size of MoS₂ transferred at low stress, resulting in a thinner, more uniform transfer film. This hypothesis was supported by the finding that sliding on Au-doped MoS₂ resulted in the formation of pure, crystalline, aligned MoS₂ about 1 nm thick on the coating [166, 167].

6 Future research direction

6.1 Expanding the role of MAX phases in tribology

MAX phases are a class of nanolaminate ternary carbides/nitrides that have gained immense interest in tribological applications due to a unique combination of metallic and ceramic properties. Incorporation of these materials into molybdenum disulfide coatings could lead to a dramatic enhancement of wear resistance and lubrication. The self-lubricating properties of MAX phases, especially under heavy loads and extreme conditions, complement the layered structure of MoS₂ and create hybrid coatings with improved thermal stability and oxidation resistance. Future research should optimize deposition techniques and investigate different MAX phase-MoS₂ combinations to tailor coatings for specific industrial applications such as aerospace and high-temperature environments. Ahmed [173] have shown that adding the MAX phase can increase the durability of MoS₂ coatings tenfold and increase oxidation resistance.

6.2 Searching for new dopants and doping techniques

While Cr, Ti, and Ag have demonstrated their effectiveness as traditional dopants in improving the tribological performance of MoS₂, new materials like V, Mo₂N, and CNTs provide bright prospects. Such dopants can reduce friction, improve wear resistance, and increase thermal stability. More advanced methods like HiPIMS or hybrid CVD techniques are encouraged, where uniform and controlled doping is essential. Furthermore, multi-element co-doping such as Carbon/Nitrogen [174] can provide previously unexploited synergistic effects and unprecedented durability and tribological properties.

6.3 Integration of two-dimensional (2D) materials

Another direction involves incorporating 2D materials, including graphene, boron nitride, and their derivatives, into MoS₂ coatings to enhance lubrication and reduce wear. These materials provide a complementary lamellar structure that allows the creation of stable lubricating films even under very harsh conditions. It is also important to understand, at a molecular level, the interactions between these 2D materials and MoS₂ to design composite coatings with properties as close to optimum as possible. Researchers are now moving toward different layered dopants, such as hexagonal boron nitride [175], WS₂ [176] to achieve lower shear modulus in solid lubricants.

6.4 Gradient and layered structure development

Gradient and multilayered structures, changing composition or switching between different materials gradually, exhibit great potential in tribological applications. For instance, such gradient structures as changing from Mo₂N to MoS₂ can enhance the substrate's hardness with excellent lubricity at the surface. Similarly, alternating layers between the MAX phases and MoS₂ combine both advantages. Advanced deposition techniques like pulsed laser deposition and atomic layer deposition need to be developed in order to fabricate these complex structures.

6.5 Sustainable environmentally coatings

With the emphasis on sustainability, future research needs to be on developing environmentally friendly coatings. That includes the investigation of bio-derived dopants, low-temperature synthesis methods, and coatings with minimal use

of hazardous substances. Besides, recycling and reusing MoS₂-based coatings may become an important area of focus, especially in industries where extensive material usage cannot be avoided.

7 Conclusion

In summary, dopants consistently improve the tribological performance of MoS₂ coatings. These dopants can be broadly categorized into hard materials (such as Ti, Ni, and Sb₂O₃) and soft materials (like Au). Regarding oxidation resistance, dopants may densify the film, limit the number of reactive sites, passivate edge sites, or sacrificially bond with oxygen, all potentially contributing to enhanced oxidation resistance of doped MoS₂ films.

Different mechanisms can improve the coatings' friction, wear, and lifespan. Hard materials densify and harden the film, directly improving wear resistance. On the other hand, soft materials may facilitate the formation of thinner and more uniform MoS₂ films on the coating's surface, indirectly reducing wear. Coatings containing both hard and soft materials, such as Sb₂O₃/Au, may benefit from both mechanisms. However, it is essential to note that wear is often inversely related to friction, especially at low temperatures.

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